

The Aggregate Impact of Policy Uncertainty in the Green Energy Transition

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Abstract

Federal subsidies for wind and solar are a central instrument of U.S. electricity-sector decarbonization, yet their availability has historically been uncertain. I study the macroeconomic and environmental costs of renewable policy uncertainty in a dynamic spatial general-equilibrium model with stochastic subsidies and irreversible generation capital. I compute counterfactual transition paths under a permanent subsidy and a budget-neutral stochastic subsidy that switches on and off. Relative to the stochastic policy, the permanent subsidy produces 13% more renewable capacity and reduces cumulative electricity-sector CO₂ emissions by 3.2% (2.29 Gt), avoiding \$321 billion in climate damages and generating a present-value welfare gain of \$208 billion. Most of the emissions difference comes from existing coal and gas plants generating more when renewable investment is delayed during subsidy lapses. The effects are largest in regions such as the Midwest, where the subsidy changes whether new renewable capacity is built and existing fossil plants can fill the generation gap during a lapse.

JEL Codes: D81, E22, H23, Q42, Q48, Q54, R13

Keywords: green energy transition, policy uncertainty, irreversible investment, renewable subsidies, spatial general equilibrium

1 Introduction

For decades, governments have relied on climate policies to incentivize decarbonization of the energy sector. The majority of the macroeconomics literature that studies the effects of climate policies such as carbon tax (Goloso et al., 2014), carbon tariffs (Casey et al., 2026a) and renewable subsidies (Arkolakis and Walsh, 2023) takes the presence of the policy as given. In practice, developers make long-lived investment decisions without knowing whether government support will remain available in the future. The U.S. has subsidized wind through federal tax credits since 1992 and solar since 2006, but that support has repeatedly changed:¹ the wind credit lapsed four times between 1999 and 2015; the 2022 Inflation Reduction Act (IRA) then extended federal clean-electricity tax credits, before the 2025 One Big Beautiful Bill Act (OBBBA) terminated the tax credits for new wind and solar projects. The aggregate cost of renewable policy uncertainty in the United States is unknown.

In this paper, I quantify the macroeconomic and environmental value of making renewable support permanent. To do this, I build a dynamic spatial general-equilibrium model with a tractable electricity sector, in which generation capital is irreversible, its investment is subject to convex adjustment costs, and the renewable subsidy is stochastic, with regions holding rational expectations about its renewal. The model distinguishes generation capacity from electricity generation: demand varies within the year, and available technologies are dispatched in merit order according to their marginal costs. This allows changes in subsidy availability to affect emissions both through the generation of the existing fossil fleet and through investment in new fossil capacity. I compute counterfactual transition paths under a permanent subsidy and a budget-neutral stochastic subsidy that switches on and off. Under the uncertain subsidy, regions invest in 13% less renewable capacity than under a permanent one and build more long-lived gas, raising cumulative CO₂ emissions by 3.2%. At the same budget, making the subsidy permanent generates an aggregate welfare gain with a present value of \$208 billion.

These aggregate results conceal substantial regional heterogeneity. Permanence has little effect where renewable projects are profitable without federal support or where they remain unprofitable even with the subsidy. The effects are largest in marginal regions where federal support changes the renewable investment decision and existing fossil capacity can increase generation when that investment is delayed. The Midwest combines both features: under a

¹I refer to the federal production and investment tax credits as renewable subsidies throughout the paper.

permanent subsidy it builds the most additional renewable capacity, while under intermittent support its existing coal and gas fleet supplies more generation when renewable investment is delayed. As a result, the Midwest also has the largest reduction in fossil generation under permanent support.

An uncertain on-and-off subsidy changes investment on two margins. First, during a subsidy lapse, a developer can build without the subsidy or wait for support to return. When the return date is uncertain, waiting has an option value: a project built during the lapse does not receive the subsidy, while one that waits can qualify when support returns. However, electricity demand continues to grow during policy lapses: after roughly fifteen years of little growth, [EIA \(2025\)](#) projects U.S. electricity demand to grow by roughly 0.9–1.6% per year through 2050, driven in part by data centers and the electrification of transportation and buildings. Growing demand is partly met by irreversible gas investment while renewable investment is delayed. Second, developers can bunch investment into periods in which the subsidy is available. Because investment is costly to accelerate due to the presence of adjustment costs, this timing response creates bunching around eligibility windows and occurs even when future policy changes are known. A permanent subsidy removes both margins by making eligibility continuously available.

I motivate these distortions with the evidence from the history of U.S. renewable tax credits. I show that the production tax credit is large enough to affect the decision of developers whether wind projects are profitable, and wind investment historically adjusts around changes in subsidy eligibility. I also use renewable-equity prices to test whether recent changes in federal support were known in advance. Renewable valuations respond sharply to the IRA agreement, the 2024 election, House passage of the OBBBA repeal, and the Senate’s subsequent revision, with little corresponding movement in fossil-energy equities. I use this suggestive evidence to build a model in which subsidy availability affects renewable investment and investors form expectations over future policy changes without knowing their realized dates in advance.

I calibrate the model using the U.S. electricity data (EIA) and the political history of the subsidy. The model has 13 electricity regions that differ in renewable resources, demand, initial generation capacity, and neighbors with whom they can trade electricity. Renewable and gas capital costs, electricity demand, fuel costs, capacity factors, and regional renewable resources are taken from external data and are exogenous in the model; investment-cost parameters are calibrated to national wind and solar capacity growth. The stochastic subsidy process is calibrated separately using the observed transitions in unified federal control.

The model reproduces untargeted aggregate emissions, wholesale electricity prices, and the generation mix.

The welfare gain from permanent support comes from avoided climate damages rather than higher private consumption. A permanent subsidy increases clean-capital investment and reduces consumption relative to the uncertain policy; this consumption cost has a present value of \$113 billion. The additional renewable capacity, however, avoids \$321 billion in climate damages. The permanent policy breaks even at a social cost of carbon of \$67 per ton of CO₂, below recent official estimates. The value of permanence is also large relative to the value of providing an uncertain subsidy at all: relative to no subsidy, the uncertain policy generates \$47 billion in net benefits.

In the counterfactuals, I decompose the emissions difference to show why distinguishing generation capacity from electricity generation is important for measuring the environmental consequences of uncertain support. Most of the additional emissions do not come from the new gas capacity induced by delayed renewable investment. While renewable investment is delayed, existing fossil plants generate more and part of the growing capacity requirement is met by new gas investment: higher usage of existing coal and gas accounts for 80% of the cumulative emissions difference. The remaining 20% comes from additional gas capacity that would not have been built under a permanent subsidy. Renewable construction can continue when the subsidy returns, but the additional gas capacity is irreversible and continues to generate potentially for decades. The emissions effect of intermittent support therefore depends not only on which technologies are built, but also on how the existing generation fleet is dispatched.

I also ask whether the value of permanence requires uncertainty about future policy dates. I solve the model along a deterministic on-and-off subsidy path whose complete future schedule is revealed to firms in 2028. Firms respond by shifting renewable investment toward known eligibility windows, producing substantial investment bunching. Permanent support remains valuable. The baseline \$208 billion should therefore be interpreted as the value of replacing stochastic, intermittent support with continuous support, rather than as the welfare cost of uncertainty about policy dates alone.

The results are robust to the main uncertainty in the policy calibration. The duration of a future subsidy lapse is not identified by the historical tax-credit record, because past lapses were short and ultimately reversed. I therefore treat the central calibration as a data-calibrated benchmark and bracket it with alternative forward regimes ranging from a

rapidly returning subsidy to permanent repeal. The value of permanence remains positive throughout, ranging from \$43 billion to \$255 billion. The result also remains positive when climate damages are discounted at the private market rate.

The model abstracts from several forces that could change the magnitude and spatial distribution of these effects. Welfare includes the environmental damages from CO₂ but not local air-pollution damages from fossil generation. Renewable capital costs follow an exogenous global cost path rather than responding endogenously to U.S. deployment through learning-by-doing. I also abstract from grid storage and state clean-electricity policies, which could alter renewable investment and its response to changes in federal support. Section 7.4 discusses how these and other omitted mechanisms affect the interpretation of the quantitative results.

1.1 Related literature

This paper contributes to the literature on policy uncertainty and irreversible investment. With sunk costs, uncertainty creates an option to wait and can delay investment (Bernanke, 1983; Dixit and Pindyck, 1994). The expectations of future policy can also produce anticipatory behavior. Lemoine (2017) shows that the anticipated carbon price affects current energy investment, while French and Vreugdenhil (2026) show that firms acquire permits before anticipated regulation to preserve the option to invest under the existing regime. Gowrisankaran et al. (2025) study how uncertainty over environmental regulation changes investment in coal generation. Closest to my paper, Fried et al. (2022) quantify the effects of uncertainty over whether a carbon tax will be adopted. In their paper, the possibility of future carbon pricing shifts investment toward clean capital before the policy arrives. I study uncertainty about the continuation of an existing renewable subsidy. This introduces the value of waiting. When a subsidy lapses, renewable developers have an incentive to wait for renewal. Due to the growing electricity demand, the missed renewable investment is partly replaced by irreversible gas capacity. A subsidy lapse can generate a fossil-capital stock that persists after support returns. I quantify the value of making stochastic, intermittent support permanent and trace the resulting effects on investment, emissions, and welfare in general equilibrium.

Another related literature studies renewable subsidies and the timing of renewable investment. The repeated expirations of the U.S. production tax credit generated pronounced boom-bust investment cycles in wind generation (Barradale, 2010). Chen (2025) studies these

investment patterns in a dynamic model of the wind industry and shows that anticipated subsidy expiration can accelerate investment as developers rush to build before eligibility ends. My model includes this anticipatory response but also studies investment during a subsidy lapse, when developers can delay investment waiting for the subsidy to return. These forces generate different timing responses depending on firms’ information. When future policy dates are uncertain, developers face an option to wait during a lapse. When the dates are known, the investment is shifted into eligibility windows. I study these effects in general equilibrium and quantify their consequences for fossil generation, investment, emissions, and welfare. More broadly, this paper relates to work on the dynamic design of renewable support. [Langer and Lemoine \(2022\)](#) characterize optimal dynamic adoption subsidies, [Aldy et al. \(2023\)](#) compare investment and production subsidies for wind generation, and [Ricks and Kay \(2025\)](#) study the design and consequences of time-limited renewable subsidies. I hold the expected subsidy budget fixed and study the gains from making stochastic on-and-off support permanent.

The paper also contributes to work on climate policy and the energy transition in general-equilibrium. General-equilibrium climate models study carbon pricing, directed technical change, and clean-energy subsidies ([Golosov et al., 2014](#); [Acemoglu et al., 2012, 2016](#); [Fried, 2018](#); [Arkolakis and Walsh, 2023](#); [Casey et al., 2026b](#)). I add a stochastic policy into a dynamic spatial general-equilibrium model that separates electricity-generation capacity from electricity generation. This distinction is important for quantifying the environmental effects which depend on how intensively existing plants are dispatched, and whether new fossil capacity is built. I combine the dynamic investment problem with merit-order dispatch and heterogeneous regional renewable resources. This builds on work showing that the value of intermittent renewables depends on when they generate and which fossil generation they displace ([Joskow, 2011](#); [Hirth, 2013](#); [Novan, 2015](#); [Gowrisankaran et al., 2016](#)). In the model, these electricity-market features determine how changes in subsidy availability translate into fossil investment, generation, and emissions.

Roadmap. The rest of the paper is organized as follows. Section 2 describes the institutional background. Section 3 presents empirical evidence on whether subsidy availability affects investment decision and whether the changes in subsidy were foreseen. Section 4 sets up the model and Section 5 calibrates it. Section 6 describes the counterfactuals, Section 7 reports the results, and Section 8 concludes.

2 Institutional background

The U.S. electricity sector has undergone a substantial change in its generation mix over the past two decades and this transition had two margins: switching from coal to natural gas, and then switching from natural gas to renewables. Coal was the leading source of electricity generation in the majority of states in 2001, but by 2025 natural gas had displaced it in most states, while wind had become the leading source in five states (Figure 1). Over the same period, wind and solar installation costs fell substantially, by 55% and 83% since 2010, respectively. Renewable capacity has consequently expanded rapidly, although fossil fuels continue to supply a large share of U.S. electricity.

The transition is geographically uneven because renewable suitability differs across the regions. Figure 2 reports wind and solar capacity factors² across the 13 electricity regions used in the model. Wind is especially suitable for the interior regions and Texas, while solar resources are strongest in the Southwest and California. These differences affect the cost of renewable electricity and therefore where wind and solar capacity is built. They also make the effect of a common federal subsidy heterogeneous across space and can create winners and losers.

The Federal Government uses separate tax credits for wind and solar: the Production Tax Credit for wind (a per-MWh payment), and the Investment Tax Credit for solar (a fixed share of capital cost). Both are claimed under a start-of-construction provision that grandfatheres the subsidy at the rate that exists when a project breaks ground³. The wind PTC lapsed repeatedly after 2000, and each lapse was a brief congressional delay and was renewed retroactively. A wind project that qualifies while the PTC is available retains the credit over its eligibility window⁴ so projects that have already qualified remain insulated from subsequent changes in eligibility. Solar subsidies were stable until 2025. The 2022 Inflation Reduction Act committed both subsidies for a decade. The 2025 One Big Beautiful Bill Act then repealed the subsidies for projects placed in service after 2027, the first full subsidy reversal in their history. (Figure 13) shows the full history of the subsidies.

For most of its history, subsidy availability did not depend on political party in power.

²The capacity factor is the ratio of actual electricity generation to the amount a facility would generate if it operated at nameplate capacity continuously.

³The wind production tax credit had a placed-in-service rule: a project qualified only if it reached commercial operation before the credit expired, until the American Taxpayer Relief Act (January 2013) replaced it with the start-of-construction rule.

⁴For the PTC, an eligible wind facility generally receives the credit on qualifying generation during its first ten years of operation.

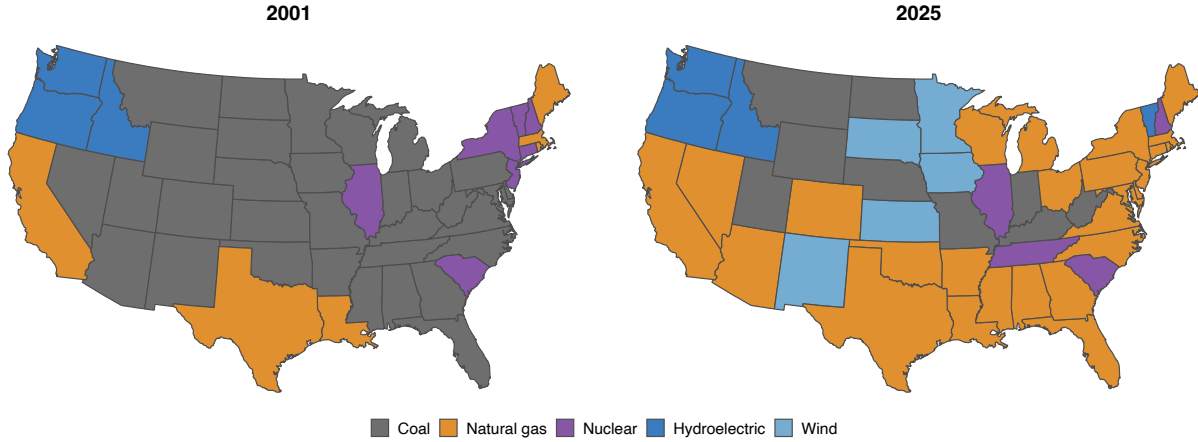


Figure 1: The top source of electricity generation by state, 2001 and 2025.
Notes. Sources: EIA Electric Power Annual, net generation by state and energy source (2001), and EIA-923 (2025).

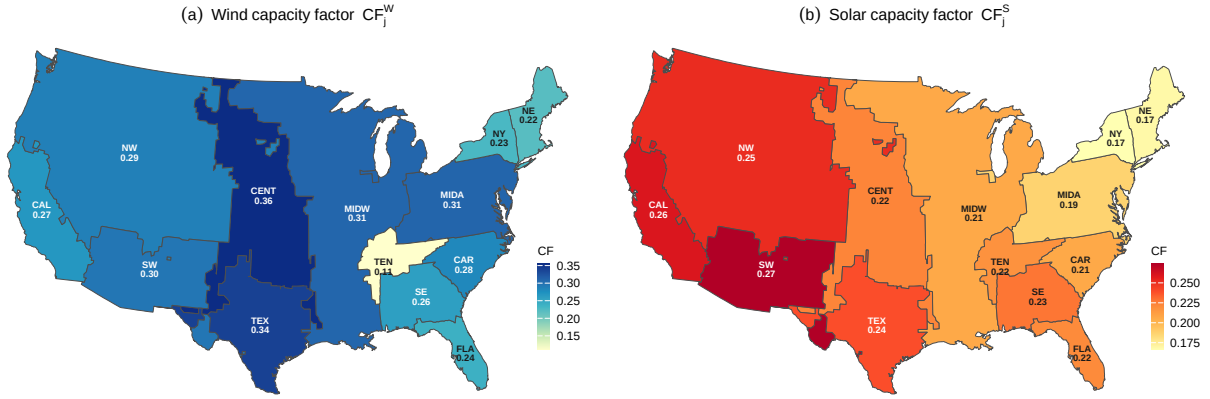


Figure 2: Wind and solar capacity factors across the 13 EIA regions.
Notes. Renewable resource geography across the 13 EIA electricity regions. *Panel (a):* wind (CF_j^W). *Panel (b):* solar (CF_j^S). Source: net generation (EIA-923) over nameplate capacity (EIA-860), 2019–2024.

Earlier lapses were short legislative interruptions followed by renewal, and the credit survived periods of unified Republican government in 2003–2006 and 2017–2018. Congressional support subsequently became increasingly polarized: Republican support for legislation containing the credit declined from roughly 75% in 1992 to zero, while Democratic support remained high. After the IRA, the credits became a partisan target, and OBBBA reversed subsidies under unified Republican control. The partisan divide also appears in public opinion: in a 2025 University of Chicago Energy Policy Institute survey, 73% of Democrats but 39% of Republicans described expanding the U.S. clean-energy industry as extremely or very

important for the federal government ([Energy Policy Institute at the University of Chicago and AP-NORC, 2025](#)).

These changes motivate modeling the relationship between future federal subsidy availability and changes in political control. Section 5 describes how I use the historical political record to calibrate the stochastic policy process.

3 Empirical evidence

I use two facts to justify the model’s assumptions about policy uncertainty. First, I show that the federal wind subsidy is large enough to affect project profitability and that wind investment adjusted around changes in subsidy eligibility in the last 25 years. Second, I use financial-market responses to recent federal policy events to test whether investors knew the realized future policy path in advance.

3.1 The subsidy availability affects wind investment

The production tax credit is large relative to wind-project revenue. A wind plant earns the wholesale price of electricity and, during its first ten years of eligibility, the \$45 production tax credit of \$26/MWh for each megawatt-hour it generates. I combine EIA-860 and EIA-923 data with engineering cost assumptions to calculate project revenue and levelized cost⁵ across the observed distribution of wind capacity factors. At the \$50/MWh average on-peak wholesale price over 2010–2024 across 8 trading hubs (EIA/ICE), the PTC accounts for 34% of project revenue during the years in which it is paid. The representative project’s levelized cost is \$53/MWh, slightly above the wholesale price of \$50/MWh without the subsidy.

Without the credit, the representative wind project breaks even at a capacity factor of 0.35; the subsidy lowers the breakeven capacity factor to 0.23. Of the 129 GW of operating U.S. wind capacity in the calculation, 48% lies above the unsubsidized breakeven threshold, 35% lies between the subsidized and unsubsidized thresholds, and the remaining 17% lies below the subsidized threshold. Thus, for a substantial share of the observed wind-capacity distribution, the PTC is large enough to move the representative project across the profitability threshold.

⁵The levelized cost of electricity is the average cost per MWh of generating electricity over a plant’s lifetime, including its up-front capital cost and operating costs.

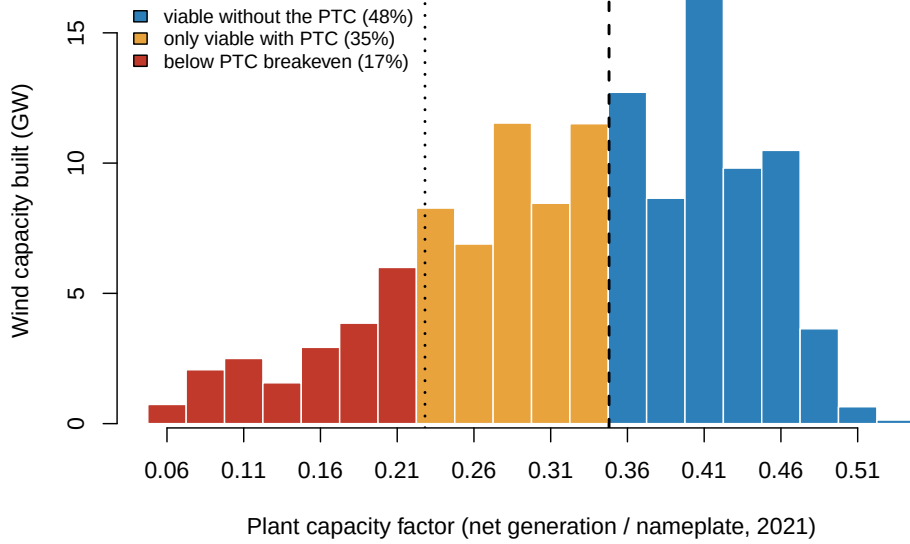


Figure 3: Wind capacity across subsidized and unsubsidized breakeven thresholds.

Notes. Operating U.S. wind capacity (EIA-860, 2021) by realized capacity factor (net generation, EIA-923, over nameplate capacity). The dashed line is the capacity factor at which the representative project breaks even at the \$50/MWh wholesale price (0.35); the dotted line is the breakeven capacity factor when the \$26/MWh credit is added (0.23). Capacity between the two lines lies between the subsidized and unsubsidized breakeven thresholds under the assumptions of the calculation. Engineering assumptions are reported in Table 11. Sources: EIA-860 and EIA-923.

Wind investment also shows pronounced boom-bust patterns around PTC expiration dates. Annual capacity additions rose before historical deadlines and fell sharply afterward. Following the four unrenewed expiration deadlines in 2000, 2002, 2004, and 2013, annual wind additions fell by 79%, 60%, 71%, and 92%, respectively (Figure 4). The largest decline followed the 2012 deadline, when additions fell from 13 GW in 2012 to 1 GW in 2013. This investment pattern around the PTC has also been documented by [Barradale \(2010\)](#) and [Chen \(2025\)](#).

3.2 Evidence on investors' information about future policy

The second set of evidence explores investors' information about future policy. In the model, I do not assume that policy changes are unexpected. Investors form rational expectations over future subsidy states and can anticipate the possibility of a policy change, but they do not know the realized future policy path. I use financial-market responses to justify this assumption.

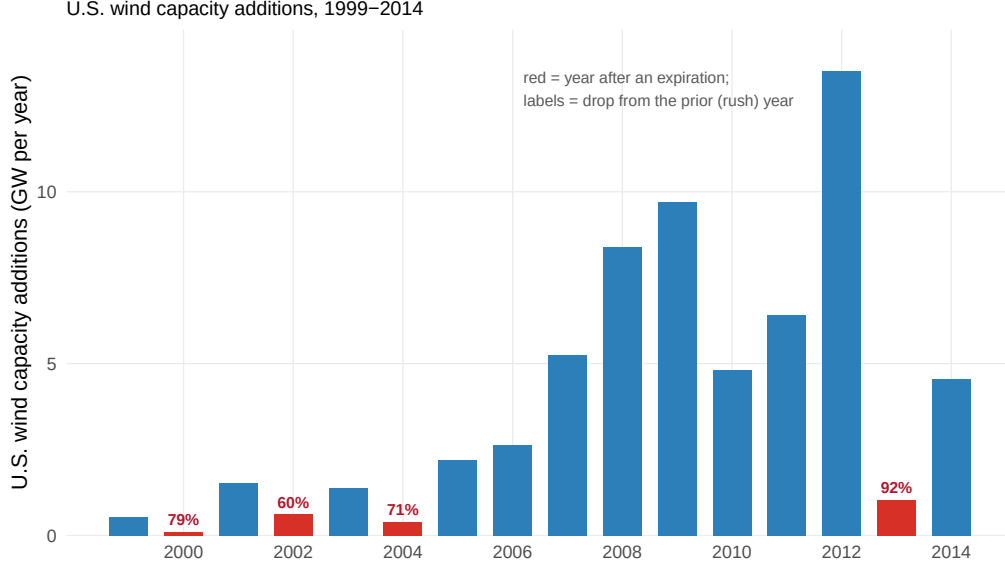


Figure 4: Wind capacity additions around the four PTC expirations.
Notes. National U.S. wind capacity additions, 1999–2014. Bars for the four years following the deadlines (2000, 2002, 2004, and 2013) are shown in red and labeled with the decline from the preceding year. Source: EIA-860.

Under full foresight, the realization of the event contains no new information and therefore should not change firm valuations. I consider four events: the Inflation Reduction Act (July 2022), the 2024 election, House passage of the OBBBA repeal (May 2025), and the Senate revision (June 2025). Figure 5 shows the changes for renewable- and fossil-energy equities. I calculate cumulative returns in excess of the S&P 500 from the event day through three trading days after the event. For each event, I summarize the differential response as

$$\Delta_e = R_e^{\text{ren}} - R_e^{\text{fos}}, \quad (3.1)$$

where R_e^{ren} and R_e^{fos} are cumulative abnormal returns for the renewable and fossil portfolios. I also report the same difference over the five trading days before each event, $[-5, -1]$, to show the extent of pre-event movement.

Renewable equities gained 12.8% following the IRA agreement, lost 20.7% after the 2024 election and 11.0% after House passage of the OBBBA repeal, and gained 9.9% when the Senate subsequently softened the repeal (Table 1 and Figure 5). The fossil portfolio moves much less around the same events. The signs of the renewable-minus-fossil differences therefore track whether an event increased or reduced expected federal support. The pre-event differences are generally smaller, although the Senate event shows substantial movement before the event window, so I do not interpret these estimates as causal event-study effects.

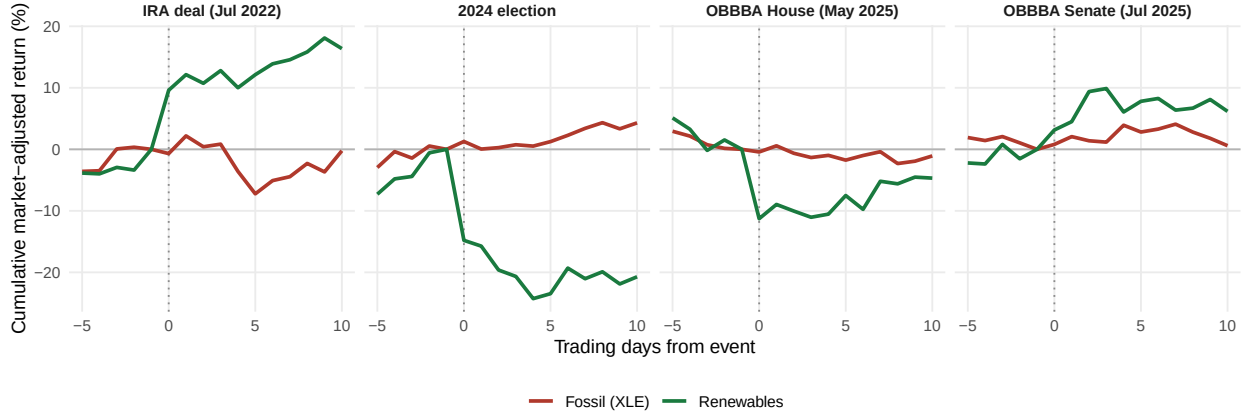


Figure 5: Stock-price reaction to federal policy events.

Notes. Cumulative return in excess of the S&P 500 normalized to zero on the trading day before the event. Source: daily stock prices.

Table 1: Renewable and fossil equity around federal policy events: cumulative abnormal returns.

Event	Renew.	Fossil	Difference $[0, +3]$		Placebo $[-5, -1]$	
	CAR (%)	CAR (%)	R - F (pp)	t	R - F (pp)	t
IRA announcement (Jul 2022)	+12.8	+0.8	+12.0	1.8	+2.2	0.3
2024 election (Nov 2024)	-20.7	+0.7	-21.4	-4.4	+3.9	0.7
OBBBA House (May 2025)	-11.0	-1.3	-9.7	-2.4	-1.7	-0.4
OBBBA Senate (Jul 2025)	+9.9	+1.2	+8.7	1.8	+11.1	2.1

Notes. Renewable basket: TAN, FAN, ICLN, First Solar, Sunrun, Enphase, and NextEra; fossil portfolio: XLE.

I use them as suggestive evidence to motivate the model’s information structure: investors can anticipate the possibility of policy changes, but they do not know the realized future subsidy path in advance.

4 Model

I develop a dynamic spatial general-equilibrium model that links stochastic renewable subsidies to investment, electricity generation, and emissions. The economy consists of 13 electricity regions connected through three interconnections. Households own generation capital and make irreversible investment decisions in wind, solar, and gas under uncertainty about future federal subsidies. Final-good firms use capital, labor, and electricity, and pro-

duction capital is traded nationally. Within each region, installed generation capacity is dispatched in merit order against fluctuating electricity demand, and regions trade electricity within their interconnection. A capacity-adequacy requirement ensures sufficient firm capacity to meet forecast demand.

4.1 Environment

Time is discrete and the horizon is infinite, $t = 2010, 2011, \dots$. There are $J = 13$ regions indexed by j , corresponding to the Energy Information Administration electricity regions and grouped into the Eastern, Western, and Texas interconnections.⁶ Each region has a representative household-generator that consumes, supplies labor inelastically, saves in a national capital market, and owns and invests in the region’s electricity-generation capital, and a representative firm that produces the final good. Regions differ in wind and solar resources (Figure 2), electricity demand, initial generation capacity, and the interconnection within which they trade.

The economy transitions from observed 2010 capital stocks toward a stochastic steady state. The transition is driven by growing electricity demand, declining global technology costs, and federal renewable subsidies.

The federal subsidy is the model’s only source of aggregate uncertainty. Subsidy availability follows a two-state Markov process $s_t \in \{0, 1\}$, where $s_t = 1$ is ”subsidy on” and $s_t = 0$ is ”subsidy off”. Household-generators know the transition probabilities but not the dates of future policy changes.

Wind earns a production tax credit (PTC) of ϕ^{PTC} per MWh. Solar gets an investment tax credit (ITC), ϕ^{ITC} , that reduces its up-front capital cost when the subsidy is active. Appendix A states the eligibility rules formally.

4.2 Households and generation investment

The representative household in region j consumes a final good C_{jt} and electricity E_{jt}^{HH} , supplies labor inelastically, saves in the national capital market, and owns the region’s gen-

⁶The 13 regions are California (CAL), Carolinas (CAR), Central (CENT), Florida (FLA), Mid-Atlantic (MIDA), Midwest (MIDW), New England (NE), Northwest (NW), New York (NY), Southeast (SE), Southwest (SW), Tennessee (TEN), and Texas (TEX).

eration capital. It maximizes

$$\mathbb{E}_0 \sum_t \beta^t \log \mathcal{U}(C_{jt}, E_{jt}^{HH}), \quad \mathcal{U}(C, E) = \left[C^{\frac{\sigma-1}{\sigma}} + \bar{\alpha}_h^{1/\sigma} E^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad (4.1)$$

where σ is the electricity-demand elasticity and $\bar{\alpha}_h$ is the electricity weight. I set σ using estimates from the empirical electricity-demand literature and, conditional on σ , calibrate $\bar{\alpha}_h$ so that electricity accounts for its observed 2% expenditure share (Section 5).

The household budget constraint is

$$C_{jt} + p_{jt}^E E_{jt}^{HH} + \sum_k \tilde{p}_t^k I_{jt}^k + \sum_k \Psi^k(I_{jt}^k, K_{jt}^k) + S_{jt} = (1 - \varsigma_t) w_{jt} L_{jt} + r_t^K A_{jt} + \Pi_{jt}^E + \mathcal{R}_{jt} + N_j T_t^{nat}, \quad (4.2)$$

where S_{jt} is saving in the national capital market, Π_{jt}^E is electricity-sector profit, $\mathcal{R}_{jt}$ is the region's lump-sum share of interregional congestion revenues, and T_t^{nat} is a national lump-sum transfer. Financial wealth evolves according to

$$A_{j,t+1} = (1 - \delta_P) A_{jt} + S_{jt}.$$

The labor-income tax ς_t finances federal subsidy payments and is non-distortionary because labor supply is inelastic.

The solar ITC lowers the effective price of solar investment,

$$\tilde{p}_t^S = (1 - \phi^{ITC} \chi_t^S) p_t^S,$$

while wind and gas pay their full capital prices, $\tilde{p}_t^W = p_t^W$ and $\tilde{p}_t^G = p_t^G$. The wind PTC instead enters electricity-sector profits as a payment on eligible wind generation. Investment is subject to convex adjustment costs $\Psi^k(I_{jt}^k, K_{jt}^k)$, and wind faces a rising regional deployment-cost curve $c_j^W(\cdot)$ as a region builds out its most productive sites. Section 5.1 gives the functional forms and calibration.

Generation capital accumulates subject to irreversibility:

$$K_{j,t+1}^k = (1 - \delta^k) K_{jt}^k + I_{jt}^k, \quad I_{jt}^k \geq 0, \quad k \in \{S, W, G\}. \quad (4.3)$$

To capture grandfathering of the wind PTC, I separately track subsidy-eligible wind

capacity:

$$K_{j,t+1}^{W,PTC} = (1 - \delta^W)(1 - h^{exit})K_{jt}^{W,PTC} + I_{jt}^W \mathbf{1}[s_t = 1]. \quad (4.4)$$

Wind built while the subsidy is active enters both the physical and subsidy-eligible stocks; wind built during a lapse enters only the physical stock. The parameter h^{exit} governs the expiration of PTC eligibility.⁷

Let

$$\mathbf{K}_{jt} = (K_{jt}^S, K_{jt}^W, K_{jt}^{W,PTC}, K_{jt}^G).$$

The value of generation capital satisfies

$$\begin{aligned} V_t(\mathbf{K}_{jt}, s_t) = \max_{I^S, I^W, I^G \geq 0} & \left\{ \Pi_{jt}^E(\mathbf{K}_{jt}, s_t) - \tilde{p}_t^S I_{jt}^S - \Xi_{jt}^W(I_{jt}^W) - \tilde{p}_t^G I_{jt}^G - \sum_k \Psi^k(I_{jt}^k, K_{jt}^k) \right. \\ & \left. + \mathbb{E}[\mathcal{M}_{t,t+1} V_{t+1}(\mathbf{K}_{j,t+1}, s_{t+1}) \mid s_t] \right\}, \end{aligned} \quad (4.5)$$

subject to (4.3)–(4.4). Here $\Xi_{jt}^W(I^W) = \int_{K_{jt}^W}^{K_{jt}^W + I^W} c_j^W(k) dk$ is wind investment expenditure along the regional site-cost curve, and

$$\mathcal{M}_{t,t+1} = \beta \frac{\mathcal{U}_C(C_{j,t+1}, E_{j,t+1}^{HH})}{\mathcal{U}_C(C_{jt}, E_{jt}^{HH})}$$

is the household's stochastic discount factor.

The wind first-order condition illustrates how expectations about future subsidy availability affect investment. For $I_{jt}^W > 0$,

$$\underbrace{c_j^W(K_{jt}^W + I_{jt}^W) + \Psi_I^W}_{\text{marginal cost}} = \underbrace{\mathbb{E}[\mathcal{M}q_{t+1}^W \mid s_t]}_{\text{energy value}} + \mathbf{1}[s_t = 1] \underbrace{\mathbb{E}[\mathcal{M}q_{t+1}^{W,PTC} \mid s_t]}_{\text{subsidy value}}. \quad (4.6)$$

When the subsidy is active, a marginal unit of wind earns both its electricity-market value and the value of the grandfathered subsidy stream.

4.3 Production firms and the national capital market

A representative firm in region j produces the final good from capital, labor, and electricity. It combines a Cobb–Douglas capital–labor bundle with electricity in a constant-returns

⁷The model approximates the statutory ten-year PTC eligibility window with an annual exit rate for eligible capacity.

CES technology,

$$Y_{jt} = Z_{jt} \left[(1 - \gamma_E)^{1/\sigma} \left(K_{P,jt}^{\hat{\alpha}_K} L_{jt}^{\hat{\alpha}_L} \right)^{\frac{\sigma-1}{\sigma}} + \gamma_E^{1/\sigma} \left(E_{jt}^{firm} \right)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad (4.7)$$

where

$$\hat{\alpha}_K = \frac{\alpha_K}{\alpha_K + \alpha_L}, \quad \hat{\alpha}_L = \frac{\alpha_L}{\alpha_K + \alpha_L}.$$

Electricity enters with the same elasticity σ as household demand. Conditional on σ , I calibrate γ_E so that electricity accounts for its observed 2% share of production costs at the reference electricity price. Cost minimization yields the firm's electricity demand below.

Production capital is supplied through a national market. Households hold financial wealth A_{jt} and firms rent production capital $K_{P,jt}$ at the common rental rate r_t^K , which clears the national capital market:

$$\sum_j K_{P,jt} = \sum_j A_{jt}. \quad (4.8)$$

4.4 The electricity sector

The household-generator owns generation capacity in hydro, nuclear, wind, solar, gas, and coal. It chooses investment only in wind, solar, and gas. Hydro and nuclear are fixed stocks, while coal follows an exogenous retirement path and no new coal is built.

Dispatch. Generation *capacity* is distinct from electricity *generation*. Figure 6 illustrates the merit-order dispatch curve.

Within-period electricity demand follows a Pareto distribution with region-specific tail index θ_j and scale $x_{jt}^{\min}$,

$$\Pr(D_{jt} > d) = \left(\frac{x_{jt}^{\min}}{d} \right)^{\theta_j}, \quad \mathbb{E}[D_{jt}] = \frac{\theta_j}{\theta_j - 1} x_{jt}^{\min}. \quad (4.9)$$

A lower θ_j corresponds to a peakier load-duration curve. Annual household and firm electricity demand is price elastic,

$$E_{jt}^{firm} = \bar{\alpha}_E Y_{jt} (p_{jt}^E)^{-\sigma}, \quad E_{jt}^{HH} = \bar{\alpha}_h C_{jt} (p_{jt}^E)^{-\sigma}, \quad (4.10)$$

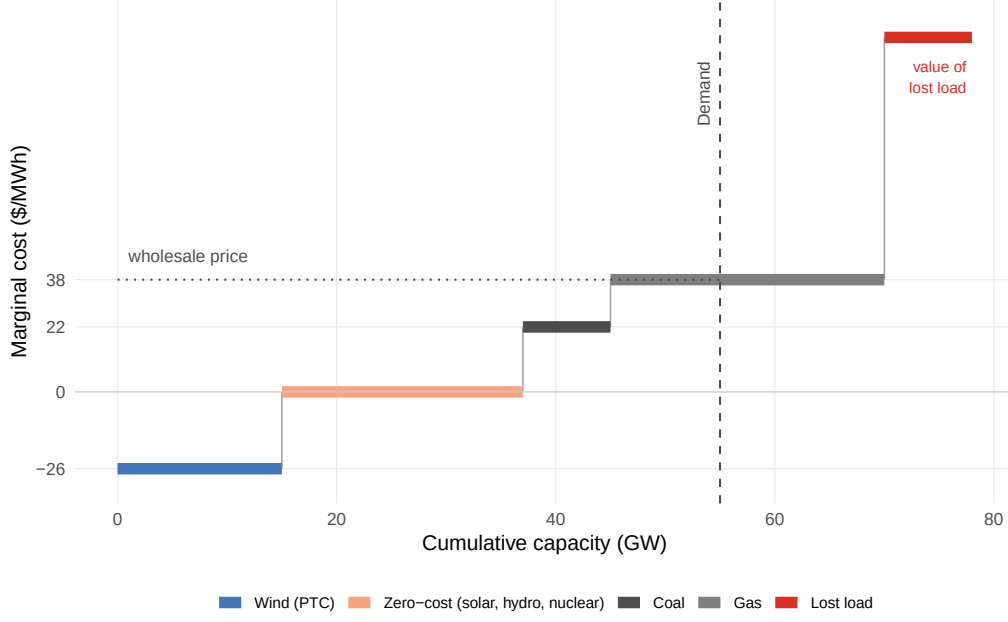


Figure 6: The merit-order dispatch curve.

Notes. Merit-order supply curve for a region in a single period. Illustrative capacities and costs.

and $x_{jt}^{\min}$ is chosen so that expected hourly demand aggregates to annual electricity demand.

The expected wholesale price is the expected marginal cost of serving load:

$$\begin{aligned} \bar{p}_{jt}^E = & mc_{jt}^{lo} \Pr(D_{jt} > Q_{jt}^0) \\ & + (mc_{jt}^{hi} - mc_{jt}^{lo}) \Pr(D_{jt} > Q_{jt}^0 + K_{jt}^{lo}) \\ & + (V_j - mc_{jt}^{hi}) \Pr(D_{jt} > \bar{Q}_{jt}), \end{aligned} \quad (4.11)$$

where Q_{jt}^0 is zero-marginal-cost capacity, mc_{jt}^{lo} and mc_{jt}^{hi} are the marginal costs of the cheaper and more expensive fossil technologies, and $\bar{Q}_{jt}$ is total capacity. Additional renewable capacity expands the zero-cost portion of the stack, makes fossil technologies marginal less often, and therefore lowers the electricity price earned by generation. Appendix A gives the corresponding closed-form expression under Pareto demand.

Capacity adequacy. I impose a reserve-margin constraint to ensure that each region has sufficient firm capacity to meet forecast demand at a given loss-of-load-probability target. Accredited capacity includes dispatchable and baseload capacity and the ELCC-weighted⁸

⁸Effective load-carrying capability (ELCC) measures how much a generator contributes to system reliability. An ELCC of 20%, for example, means that 1 MW of installed capacity contributes 0.2 MW toward

contribution of wind and solar. Because gas is the endogenous source of firm capacity, the constraint places a floor on gas investment when existing accredited capacity is insufficient. Its multiplier is a capacity price p_{jt}^{cap} , the shadow value of relaxing the reserve constraint. I size each region's reserve requirement to its own exogenous forecast demand. Appendix A.3 defines accredited capacity, the reserve constraint, and the associated capacity price formally.

Trade. Regions trade electricity within, but not across, the three interconnections. Let $X_{jt} = G_{jt} - \bar{E}_{jt}$ denote region j 's net exports. The local electricity price is the interconnection hub price net of a congestion basis,

$$p_{jt}^E = \bar{p}_{mt}^E - \gamma \frac{X_{jt}}{\bar{T}L_j}, \quad (4.12)$$

where $\bar{T}L_j$ measures transmission capacity. An exporter's local price falls relative to the hub price as congestion increases, while an importer pays a higher local price. Each interconnection's hub price clears aggregate net trade among its member regions. Transmission costs are real resource costs; congestion-basis payments are transfers and are rebated lump-sum through $\mathcal{R}_{jt}$. Appendix A gives the complete electricity-sector profit and trade accounting.

National electricity-sector CO₂ emissions are

$$E_t = \sum_j (\xi^G gen_{jt}^G + \xi^C gen_{jt}^C). \quad (4.13)$$

4.5 Equilibrium

Given the exogenous paths and policy transition matrix Π , a recursive competitive equilibrium is a sequence of household allocations, generation and production capital, investment policies, electricity generation and trade, and prices such that: *(i)* households optimize consumption, saving, and generation investment; *(ii)* final-good firms maximize profits; *(iii)* generation is dispatched in merit order and electricity markets clear within each interconnection; *(iv)* the national capital market and regional labor markets clear; *(v)* the federal budget and aggregate resource constraint hold; and *(vi)* expectations are consistent with the policy transition matrix Π . The economy converges to a terminal stochastic steady state. Appendix A states the equilibrium conditions formally, and Appendix B describes the computational solution.

the capacity-adequacy requirement.

4.6 Welfare

I measure welfare as the present value of real consumption and the social cost of power-sector CO₂ emissions,

$$W^C = \sum_t \beta^t \sum_j C_{jt}, \quad W^E = \sum_t \left(\frac{1}{1+r_c} \right)^{t-2010} \text{SCC}_t E_t, \quad W^\$ = W^C - W^E. \quad (4.14)$$

The consumption component is discounted at the household’s market rate β . The climate component values each year’s emissions using the EPA social cost of carbon and discounts damages to the base year at the social rate used in its construction ([U.S. Environmental Protection Agency, 2023](#)).

Subsidies, the taxes that finance them, household electricity payments, and congestion rebates are transfers and therefore cancel from aggregate welfare. Generation investment, adjustment costs, fossil fuel use, transmission, and production-capital accumulation are real resource costs. Appendix [A.5](#) gives the equivalent aggregate resource constraint.

5 Calibration

I calibrate the model in three steps. First, I set technology, demand, emissions, and regional electricity-market parameters using external data. Second, I calibrate the investment and site-cost parameters to reproduce national U.S. wind and solar capacity levels over 2010–2024. Third, I calibrate firms’ expectations about future subsidy availability using the historical policy record and transitions in federal political control. Tables [2](#) and [3](#) summarize the data sources and parameter values.

5.1 External parameters and exogenous paths

The exogenous paths come from standard energy datasets. Wind, solar, and gas capital costs follow the NREL Annual Technology Baseline (Moderate) projection, joined to historical overnight costs from LBNL and EIA. Electricity demand follows the EIA Annual Energy Outlook 2025 reference case. Fuel costs and the exogenous coal-retirement path come from EIA. Appendix Figure [12](#) plots these paths.

Regional renewable productivity is measured using realized capacity factors, calculated as net generation from EIA-923 divided by nameplate capacity from EIA-860 and averaged over 2019–2024. I use NREL developable-resource data to calibrate the regional scale of wind and solar site availability. I use EIA-930 to calculate the shape of within-year electricity demand and calibrate the parameters for interregional trade. Appendix E describes the construction of each input and Table 10 reports regional summary statistics.

The remaining externally calibrated parameters come from engineering estimates and the empirical literature. Technology-specific depreciation rates and renewable capacity credits come from NREL, emissions factors from EPA eGRID, and production cost shares from BEA. The wind PTC and solar ITC are set to their statutory values. Panel A of Table 3 reports the full set of externally calibrated parameters.

Table 2: Data sources.

Dataset	Source	Years	Use in the model
Existing capacity	EIA-860	2010–2024	Initial stocks $K_{j,0}^k$, coal-retirement path
Generation & fuel	EIA-923	2010–2024	Capacity factors CF_j^k ; fuel marginal costs
Hourly demand & interchange	EIA-930	2021–2023	Pareto demand shape θ_j ; trade capacity $\overline{TL}_j$
Electricity demand path	EIA AEO 2025 Ref.	2024–2050	Demand growth (electrification growth)
Capital-cost projection	NREL ATB 2024 (Moderate)	2025–2050	Renewable and gas cost decline
Developable resource	NREL land-based resource	2023	Site-cost scales $\bar{K}_j^k$
Wholesale prices	ICE, NYISO	2021–2025	Interconnection trade slope s_j
Emissions factors	EPA eGRID	2022	ξ_G, ξ_C
Household counts	U.S. Census	2020	Regional L_j ; state-to-region aggregation
Policy history	PTC/ITC statutes, IRS notices	1992–2025	Eligibility; forward Markov chain
Region geography	EIA / Census TIGER	2019	13-region map

Notes. The AEO 2025 demand path and the NREL ATB 2024 cost projection are published through 2050 and held (or extrapolated along their terminal trend) over the remaining model horizon.

Table 3: Model parameters.

Parameter	Symbol	Value
<i>Panel A. Externally calibrated (data, engineering, tax code)</i>		
Discount factor	β	0.9615 ($\rho \approx 4\%$)
Wind, solar depreciation	δ^W, δ^S	0.04, 0.033 (NREL ATB)
Gas, production-capital depreciation	δ^G, δ_P	0.03, 0.05
PTC value	ϕ^{PTC}	\$26/MWh (§45)
ITC rate	ϕ^{ITC}	30% (§48)
Solar capital cost path	p_t^S	\$4.27M→\$0.79M/MW
Wind capital cost path	p_t^W	\$2.46M→\$1.10M/MW
Gas capital cost path	p_t^G	\$1.33M→\$0.74M/MW
Demand price elasticity	σ	0.3 (empirical 0.2–0.4)
Emissions factors	ξ^G, ξ^C	0.40, 0.95 tCO ₂ /MWh (eGRID)
Social cost of carbon	SCC	\$190/tCO ₂ (EPA 2023)
PTC vintage duration	h^{exit}	0.1 (§45)
Site-cost curvature	η	2 (NREL supply curves)
Capacity factors	CF_j^k	wind 0.04–0.38, solar 0.19–0.32
Trade slope (congestion)	$s_j = \gamma/\sqrt{TL_j}$	0.24–8.6, $\gamma \approx 2$ (ICE/NYISO, EIA-930)
Reliability target (LOLP)	ε	0.002 (NERC 1-in-10)
Capacity credit (ELCC)	ζ^W, ζ^S	0.15, 0.10 (NREL)
Cost shares (capital/labor/electricity)	$\alpha_K, \alpha_L, \alpha_E$	0.33, 0.65, 0.02 (BEA)
Electricity demand weights	$\bar{\alpha}_h, \bar{\alpha}_E$	$0.0014 = 0.02 \times 45^{-0.7}$ (2% share)
<i>Panel B. Internally calibrated to the 2010–2024 capacity growth</i>		
Wind, solar adjustment cost	$\bar{\psi}_W, \bar{\psi}_S$	2×10^5 each
Gas adjustment cost	$\bar{\psi}_G$	10^7 (combined-cycle builds more slowly)
Wind site-cost scale	κ_W	0.08 (2024 wind ≈ 143 GW)
Solar site-cost scale	κ_S	0.22 (2024 solar ≈ 136 GW)

Notes. Panel A is set from external data, engineering studies, and the tax code. Panel B’s four deployment-cost parameters are calibrated to the observed 2010–2024 renewable capacity growth, matching the 2024 capacity levels. The policy Markov chain is calibrated separately (Section 5.3).

5.2 Internal calibration

Four parameters govern how quickly renewable investment responds to changes in profitability. Investment adjustment costs are quadratic in the investment rate,

$$\Psi^k(I, K) = \frac{\psi^k}{2} \left(\frac{I}{K} \right)^2 K,$$

and capture the increasing cost of expanding generation capacity rapidly. Wind deployment costs also rise as a region exhausts its most productive sites:

$$c_j^W(K_{jt}^W) = \tilde{p}_t^W \left[1 + \left(\frac{K_{jt}^W}{\bar{K}_j^W} \right)^\eta \right], \quad (5.1)$$

where $\bar{K}_j^W$ is proportional to region j 's developable wind resource and I set $\eta = 2$.

I calibrate the wind and solar adjustment-cost parameters and the wind and solar site-cost scales jointly to reproduce national wind and solar capacity levels over 2010–2024. Adjustment costs determine how quickly investment can respond when profitability changes. The site-cost scales determine how rapidly increasingly expensive locations are reached as cumulative deployment rises.

5.3 Policy expectations

The availability of the subsidies follows a two-state Markov process,

$$s_t \in \{0, 1\},$$

with π_{11} denoting the probability that an active subsidy remains active and π_{00} the probability that a lapse continues for another year.

Before 2025 subsidy lapses were brief and ultimately reversed, and there is no historical episode of a long repeal. I use the history of federal political control to discipline π_{00} . I classify each Congress since 1993 as a Democratic trifecta, a Republican trifecta, or divided government and estimate transitions among the three states (Table 9). I map a Democratic trifecta into the active subsidy state and a Republican trifecta into the lapsed state.

I convert the estimated political process to annual transition probabilities by preserving its expected spell lengths. This gives $\pi_{11} = 0.90$ (an expected ten years active) and $\pi_{00} = 0.94$ (an expected seventeen-year lapse). The robustness analysis also considers faster renewal and permanent repeal. Section 7 shows how the results change across this range.

5.4 Validation: untargeted moments

The calibration uses national wind and solar capacity levels from 2010–2024, but does not target emissions, generation shares, electricity prices, the location of renewable capacity, or electricity trade.

At the national level, the model produces 1.52 Gt of power-sector CO₂ emissions in 2024, compared with 1.5 Gt in the data. Wind supplies 10% of generation in both the model and the data, hydro and nuclear supply 25% in the model and 24% in the data. Fossil fuel together supply 57% and 58%, respectively. The main difference is solar: its share is 8% in the model and 5% in the data. The model’s average wholesale electricity price also falls within the observed 2021–2024 range.

The regional allocation of renewable capacity is not a calibration target. However, the Spearman rank correlation between modeled and observed wind capacity across the 13 regions in 2024 is $\rho = 0.96$. The match is weaker for solar, with $\rho = 0.57$. The model gets the direction of net electricity trade right in 8 of the 12 regions that trade across regional boundaries (excluding Texas as it does not trade).

Figures 14 and 15 provide the corresponding dynamic and regional comparisons.

Table 4: Untargeted validation: model versus data.

Moment	Model	Data
<i>Panel A. Aggregate moments</i>		
Power-sector CO ₂ , 2010 (Gt/yr)	2.15	2.2
Power-sector CO ₂ , 2024 (Gt/yr)	1.52	1.5
Wholesale price, 2021–2024 (\$/MWh)	41	35–45
Wind generation share, 2024 (%)	10	10
Solar generation share, 2024 (%)	8	5
Baseload generation share, 2024 (%)	25	24
Fossil generation share, 2024 (%)	57	58
<i>Panel B. Spatial agreement</i>		
Wind capacity across regions, Spearman ρ	0.96	
Solar capacity across regions, Spearman ρ	0.57	
Net-trade sign	8/12 regions	

Notes. Regional capacity statistics report the Spearman rank correlation between modeled and observed 2024 capacity across the 13 regions. The trade statistic reports the number of regions for which the model reproduces the observed sign of net trade; Texas is excluded because ERCOT has no inter-connection trade in the model.

6 Counterfactuals

All counterfactuals share the realized 2010–2027 policy history and therefore enter 2028 with identical state variables.⁹ The policy experiments branch in 2028, when the OBBBA repeal takes effect.

I consider three main policy regimes: a permanent subsidy, an uncertain subsidy, and no subsidy.

6.1 Policy regimes

Permanent subsidy. The subsidy remains available in every period, $s_t = 1$. Wind receives the production tax credit of \$26/MWh and solar receives the 30% investment tax credit. Firms know that the subsidy will remain available when making investment decisions. The present value of government spending under this policy is \$231 billion.

⁹The model transition begins in 2010 and is calibrated and validated against the observed electricity transition. The 2010 start date does not affect differences across the post-2028 policy experiments.

Uncertain subsidy. The subsidy availability follows the two-state Markov process calibrated in Section 5.3. The transition probabilities are known but the dates of future lapses and renewals are not. I keep the solar ITC at 30% and set the active-state wind PTC to \$36.4/MWh, so that expected present-value subsidy spending under the uncertain policy also equals \$231 billion.

No subsidy. The subsidy is permanently unavailable, $s_t = 0$, and firms know that it will not return. This counterfactual has zero subsidy expenditure and is therefore not budget-neutral with the permanent and uncertain regimes. I use it to measure the value of providing federal support at all.

6.2 Counterfactual comparisons

My main welfare comparison is

$$W^{\text{perm}} - W^{\text{unc}},$$

which I call the *value of permanence*. The permanent and uncertain policies have the same expected present-value subsidy spending.

I also report

$$W^{\text{unc}} - W^{\text{none}},$$

the value of the uncertain subsidy relative to no federal subsidy. The two differences add up to the value of permanent support relative to no subsidy:

$$W^{\text{perm}} - W^{\text{none}} = (W^{\text{perm}} - W^{\text{unc}}) + (W^{\text{unc}} - W^{\text{none}}).$$

6.3 Perfect-foresight counterfactuals

I run two additional counterfactuals in which the complete future subsidy schedule is known in 2028.

In the *short-lapse* known subsidy schedule scenario, the subsidy is off from 2028 through 2033, on from 2034 through 2043, and off again beginning in 2044. In the *long-lapse* known

subsidy schedule scenario, the subsidy is off from 2028 through 2044, on from 2045 through 2054, and off again beginning in 2055. Household-generators know the relevant schedule in 2028 and solve their investment problem using that known sequence rather than expectations over future subsidy states.

I adjust the active-state wind PTC in each foresight experiment so that present-value subsidy spending is equal to spending under permanent support. The permanent policy spends \$230.9 billion in present value, compared with \$232.0 billion under the short-lapse path and \$232.3 billion under the long-lapse path. The foresight experiments therefore compare continuous and unstable support at essentially the same fiscal cost while removing uncertainty about future policy dates.

7 Results

7.1 The value of a permanent subsidy at a fixed budget

At the same expected budget, a permanent subsidy produces more renewable capacity and less gas than the stochastic on-and-off subsidy. The permanent policy has 137 GW more renewable capacity and 37 GW less gas (Figure 7) by 2070. The renewable-capacity gap is not monotone: it widens to approximately 156 GW around 2062 and then begins to narrow as the subsidy returns under the uncertain policy and renewable investment resumes. The gas response is more persistent: capacity built during a lapse is irreversible and remains in the generation stock until it depreciates.

The capacity response can be translated into higher fossil generation under the uncertain policy (Figure 8). Cumulatively, the stochastic on-and-off policy produces 2.29 Gt more CO₂ through 2070, or 3.2% relative to the permanent policy, which is equivalent to roughly a year and a half of U.S. power-sector emissions in 2024.

Most of the emissions difference does not come from the newly added gas capacity itself. Existing fossil plants run more when renewable investment is delayed. Additional generation from existing coal and gas accounts for 1.83 Gt, or 80% of the cumulative emissions gap. Of this component, 52% comes from existing gas and 48% from coal. The remaining 0.46 Gt (20%) comes from gas capacity that is built under the uncertain policy but not under the permanent one.

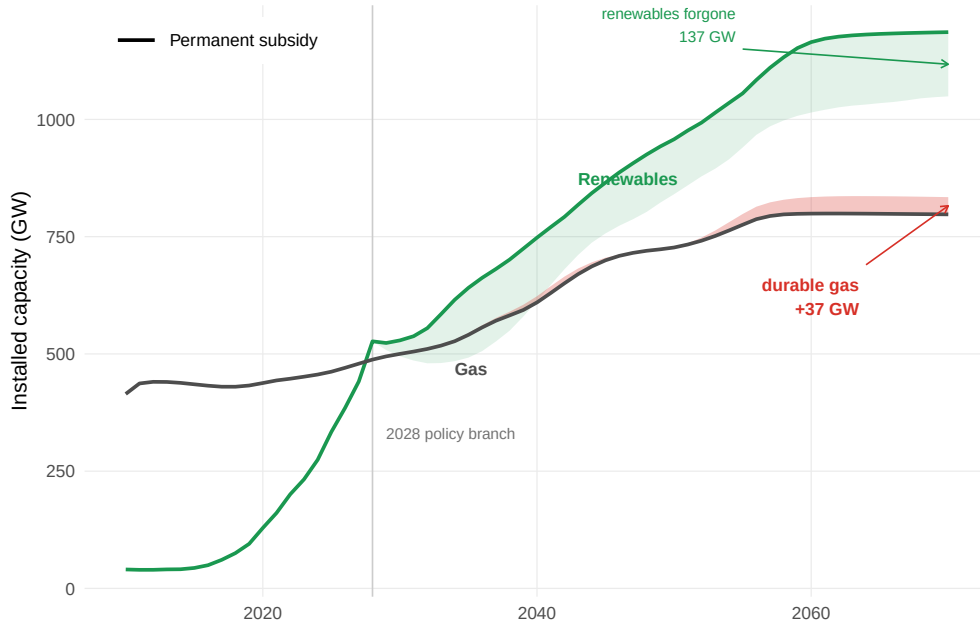


Figure 7: **Renewable and gas capacity under a permanent versus a budget-neutral uncertain subsidy.**

Notes. Installed renewable (wind plus solar) and gas capacity under a permanent subsidy (solid) and a budget-neutral uncertain subsidy (dashed).

Table 5: The value of a permanent subsidy, decomposed.

(\$B, present value)	Permanent – uncertain	Uncertain – no subsidy
Climate gains	321	181
Avoided emissions, existing coal	158	85
Avoided emissions, existing gas	104	56
Avoided emissions, new gas	59	40
Consumption	–113	–134
Renewable & baseload income	144	66
Wages	16	12
Gas & coal profit	–23	–9
Generation investment	–251	–203
Value	208	47

Notes. Present value in billions of dollars. Climate gains value avoided CO₂ at the EPA year-specific social cost of carbon discounted at 2%, split by the source of the avoided emissions.

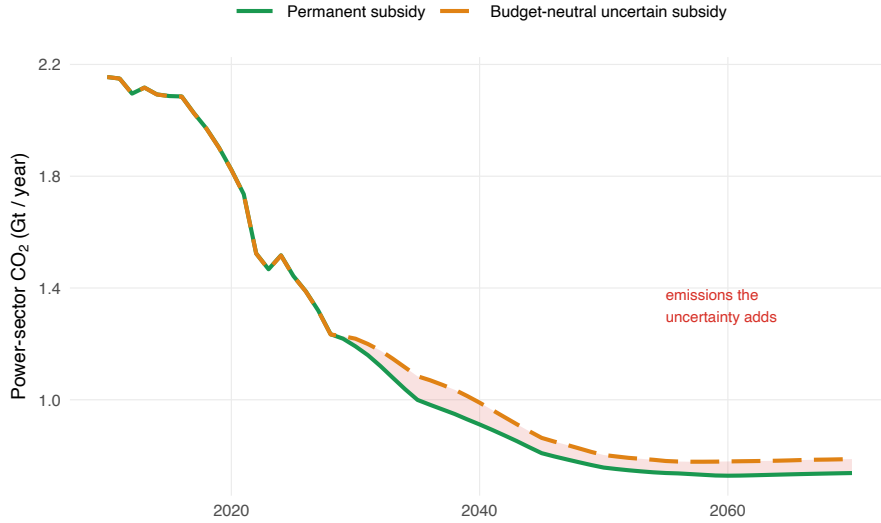


Figure 8: Annual power-sector CO₂ by policy regime.

Notes. Annual U.S. power-sector CO₂ emissions under a permanent subsidy and a budget-neutral uncertain subsidy.

Table 5 shows welfare changes. Replacing the uncertain subsidy with a budget-neutral permanent one raises welfare by \$208 billion in present value. The gain comes from avoided climate damages rather than higher private consumption. Lower emissions are worth \$321 billion at the baseline social cost of carbon, with most of the benefit coming from lower generation by existing coal and gas and the remainder from gas capacity that is never built.

Permanent support has a private resource cost. It creates more renewable investment and therefore lowers consumption-equivalent welfare by 0.13%, with a present-value consumption cost of \$113 billion. The gross investment response is larger than this net consumption cost because part of the additional spending returns to households through generation income and wages (Table 5). The climate benefit is substantially larger, leaving the \$208 billion net welfare gain. The permanent policy breaks even at a social cost of carbon of \$67/tCO₂.

For comparison, the uncertain subsidy itself is worth \$47 billion relative to no subsidy, while a permanent subsidy is worth \$255 billion relative to no subsidy. The difference between permanent and uncertain support is economically large even after expected subsidy spending is held fixed. The value of permanence remains positive under faster subsidy renewal, permanent repeal, alternative reporting horizons, and discounting climate damages at the market rate; Appendix G reports these exercises.

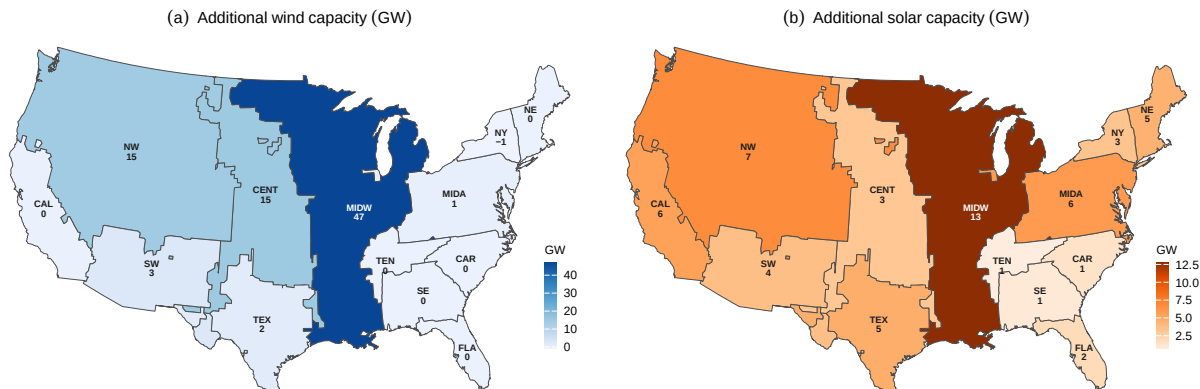


Figure 9: Regional capacity response to a permanent subsidy.

Notes. Additional installed capacity under a permanent subsidy relative to the budget-neutral uncertain subsidy in 2070, by EIA region. Panel (a) reports wind and panel (b) solar.

7.2 Regional heterogeneity

The aggregate investment response is unevenly distributed across regions (Figure 9). The wind response is concentrated in the interior: the Midwest, Central, and Northwest together build 76 GW more wind under permanent support. In contrast, regions with poor wind resources, such as Tennessee and Florida, build little additional wind.

The solar response is also substantial, but it is more geographically dispersed. Permanent subsidy increases solar capacity across nearly every region, with the largest increase in the Midwest and significant responses in the Northwest, California, the Mid-Atlantic, New England, and Texas. The Midwest has the largest additional wind and solar capacity under permanent support.

The response is therefore not simply increasing in renewable quality. Where renewable projects are profitable without federal support, investment can continue during a lapse. At the other extreme, where renewable resources are poor, even the subsidy may not be sufficient to make additional projects profitable. Permanence matters most between these cases, where continued subsidy availability changes the marginal investment decision.

The emissions response also depends on the generation capacity already in place when the policies diverge. When renewable investment is delayed, a region with substantial existing fossil capacity can meet more electricity demand by running those plants more intensively. The emissions effect is therefore particularly large where renewable investment is marginal and an existing fossil fleet is available to fill the generation gap. The Midwest combines both

features: federal support affects marginal wind investment, while its existing coal and gas capacity provides a fossil substitute when that investment is delayed.

Electricity trade adds another spatial margin by separating the location of renewable investment from the location of the fossil generation it displaces. Regions trade within the Eastern and Western interconnections, while Texas forms a separate interconnection. Additional renewable capacity in one region can therefore reduce fossil generation elsewhere in the same interconnection.

Table 6: The value of a permanent subsidy by region, decomposed (permanent versus budget-neutral uncertain).

Region	Climate gains				Consumption					Value
	Coal	Ex. gas	New gas	Total	Invest.	Income	Wages	Profit	Total	
Midwest	88	15	12	115	-59	66	-3	-4	0	115
California	0	41	7	48	-24	10	4	-7	-17	31
Texas	34	7	0	41	-76	72	-8	1	-12	29
Mid-Atlantic	0	24	3	26	-16	7	0	0	-9	17
Tennessee	0	4	0	4	7	1	0	0	8	11
New England	0	9	7	16	-12	6	0	-1	-7	9
Florida	1	12	0	13	-8	3	0	0	-5	8
New York	0	7	2	10	-8	4	1	-2	-5	4
Carolinas	6	0	0	6	-3	0	1	0	-2	4
Southeast	1	3	0	3	-2	1	0	0	-2	2
Southwest	7	2	1	10	-13	-3	4	-2	-14	-4
Northwest	18	-4	7	21	-19	-13	13	-7	-26	-4
Central	4	5	0	9	-17	-9	4	0	-23	-14
National	158	104	59	321	-251	144	16	-23	-113	208

Notes. Present value in billions of dollars, by region, sorted by value. Climate gains value the region's own avoided CO₂ at the EPA year-specific social cost of carbon discounted at 2%, split by source (existing coal, existing gas, new gas). Consumption is the change in the present value of the region's consumption; income, wages, and profit raise it, generation investment enters negatively. Value = climate gains + consumption. Rows sum to the National row.

California illustrates this mechanism. California is a net importer, and its own renewable-capacity response is relatively small. Under the permanent subsidy, however, the Northwest and Southwest build more renewable capacity and supply more clean electricity within the Western Interconnection. California imports part of this additional generation and runs its own gas capacity less. The location of renewable investment therefore differs from the location of some of the displaced fossil generation.

Table 6 reports the corresponding regional decomposition. I assign the climate component according to the region in which power-sector CO₂ emissions change. Because CO₂ damages are global, this allocation is an accounting decomposition of the national climate benefit

rather than a measure of local environmental benefits or regional welfare incidence. The model does not include damages from local air pollutants.

Most of the Midwest climate benefit comes from lower utilization of fossil capacity that already exists. Of its \$115 billion in allocated climate gains, \$88 billion comes from reduced generation by existing coal and another \$15 billion from existing gas; only \$12 billion comes from avoiding new gas capacity.

7.3 Permanence under perfect foresight

Figure 10 shows the investment response under the short-lapse path. Renewable investment falls during the 2028–2033 lapse, jumps when eligibility returns in 2034, and rises again before the known 2044 lapse. Household-generators use their knowledge of future policy dates to shift construction toward periods in which projects can qualify for federal support.

The response is not simply a delay until the subsidy returns. Renewable construction becomes concentrated inside the known eligibility window because regions also know when the next lapse will begin. Investment under perfect foresight temporarily exceeds investment under permanent support, so the renewable-capacity gap changes sign over the transition. Shifting investment across years is costly because investment is subject to convex adjustment costs, while electricity demand continues to grow during the lapse. Substantial capacity differences therefore remain even though the policy dates are known.

Table 7 compares the two foresight paths. The short-lapse policy spends \$232.0 billion in present value, compared with \$230.9 billion under permanent support, yet permanence is worth \$245 billion. Removing uncertainty about the policy dates therefore does not eliminate the value of budget-neutral continuous support.

Table 7: Permanent support versus deterministic perfect-foresight policies.

Policy	First lapse (years)	PV subsidy spending (\$ billion)	Value of permanence (\$ billion)
Permanent subsidy	—	230.9	—
Perfect foresight: short lapse	6	232.0	245
Perfect foresight: long lapse	17	232.3	365

Notes. In the short-lapse path the subsidy is off in 2028–2033, on in 2034–2043, and off beginning in 2044. In the long-lapse path it is off in 2028–2044, on in 2045–2054, and off beginning in 2055.

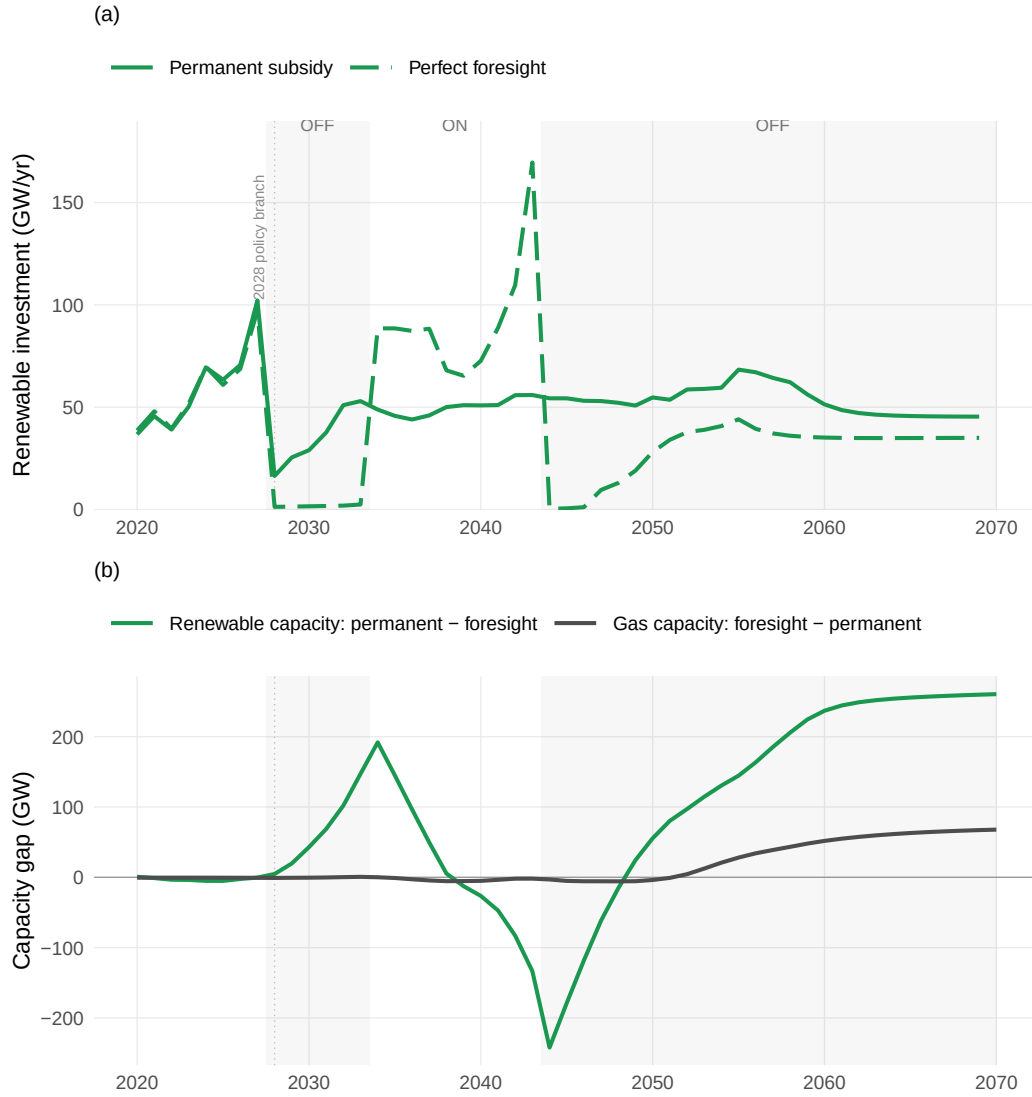


Figure 10: Investment and capacity under perfect foresight: short lapse.
Notes. Firms learn the complete future subsidy schedule in 2028. The subsidy is off in 2028–2033, on in 2034–2043, and off beginning in 2044. Panel (a) reports annual renewable investment under permanent support and the deterministic perfect-foresight policy. Panel (b) reports the renewable-capacity gap as permanent minus perfect foresight and the gas-capacity gap as perfect foresight minus permanent. Shaded areas indicate periods in which the subsidy is unavailable.

Lapse duration matters even when the policy path is known. Extending the first lapse from six to seventeen years raises the value of permanence from \$245 billion to \$365 billion. A short anticipated lapse gives firms more scope to shift investment across policy dates and to rely on subsidy-eligible wind that was grandfathered before the lapse. A longer interruption leaves less scope for this intertemporal substitution: the lapse extends beyond the ten-year PTC eligibility window, renewable capacity remains lower for longer, and fossil generation fills more of the gap.

7.4 Discussion

The model abstracts from several features of electricity sector and energy transition that could affect the magnitude and spatial distribution of the results. I discuss how each omission affects the interpretation and magnitudes of the counterfactuals.

Local air pollution. Welfare includes the climate damages from CO₂ emissions but not the local health damages from natural gas and coal generation. Lower coal and gas generation under a permanent subsidy would also reduce emissions of local pollutants such as SO₂, NO_x, and particulate matter. Quantifying these benefits would require modeling the location of marginal fossil generation that is not being used. The welfare estimates should therefore be interpreted as excluding local air-quality benefits.

Learning-by-doing. Capital costs fall along an exogenous cost curve (Section 5.1), not through endogenous learning. Location-specific learning would widen the permanence gap: a permanent subsidy deploys more, learns faster, and deploys still more, while an uncertain subsidy deploys less and stays costlier. However, because learning in wind and solar is largely global, set by world manufacturing more than by U.S. installation, this amplification is likely small. I read my results as conservative due to this omission.

Storage. The model has no grid storage. Storage complements renewables by raising the price they earn: it charges when the wholesale price is low and discharges when it is high, reducing the cannibalization that drives the price in the model, though it leaves the capacity factor unchanged. Including it would change the mechanism in two opposite ways. Some regions that did not invest during a lapse without storage would find renewables worth building without the subsidy once storage raises their value, and would build them instead of gas, narrowing the gap between the permanent and uncertain paths. But in regions where investment would still stop during a lapse, the forgone renewables would now be more

valuable, widening the gap. The net is ambiguous.

Intraday renewable generation. Wind and solar generation are represented using annual capacity factors rather than technology-specific hourly generation profiles. As a result, the model does not capture the covariance between renewable output and electricity demand. Hourly scope would increase the value of solar where its generation coincides with peak demand, reduce it as midday solar generation saturates the market, and interact importantly with storage. This incorporation would refine the technology mix and dispatch response, although the central investment mechanism would remain.

State clean-electricity policies. The model abstracts from state renewable portfolio standards and clean-electricity standards. Because state policies differ substantially across regions, incorporating them could also change the spatial distribution of the response.

8 Conclusion

Federal renewable subsidies are designed to encourage long-lived clean-energy investment, but the availability of that support has historically been uncertain. This paper studies the aggregate consequences of making federal renewable support permanent rather than delivering it through a stochastic policy that switches on and off. I develop a dynamic spatial general-equilibrium model in which wind, solar, and gas investment is irreversible, electricity is dispatched in merit order, and renewable subsidies follow a stochastic policy process. I compare a permanent subsidy with a budget-neutral stochastic subsidy that switches on and off.

I find that relative to the budget-neutral stochastic subsidy, permanent support produces 137 GW more renewable capacity and 37 GW less gas capacity by 2070. It reduces cumulative power-sector CO₂ emissions by 2.29 Gt, or 3.2%, and generates a present-value welfare gain of \$208 billion.

The emissions response operates through both generation and investment: while renewable investment is delayed during a lapse, existing coal and gas plants generate more. This margin accounts for 80% of the cumulative emissions difference. The remaining 20% comes from additional gas capacity. Renewable construction increases again when the subsidy returns, but gas built during the lapse is irreversible and remains in the generation stock until it depreciates.

The perfect-foresight counterfactual shows that this result is not driven solely by uncertainty about policy dates: even when future lapses are known and fiscal costs are held fixed, permanence remains valuable, and its value rises substantially with the duration of the anticipated lapse.

Several extensions could quantify these forces more broadly. The model isolates federal renewable subsidies and abstracts from state renewable and clean-electricity policies. Incorporating state policies would show how state support substitutes for or reinforces changes in federal policy and could alter the regional distribution of the effects. Alternative paths for electricity demand and technology costs could change both the speed of the transition and the set of regions in which renewable investment is marginal. Allowing endogenous investment in nuclear and hydroelectric generation, transmission, and storage would expand the set of technologies available when renewable support changes. Finally, a richer model of political policy formation could jointly determine subsidy changes and firms' expectations rather than taking the stochastic policy process as exogenous.

More broadly, the results point to a distinction between the size of a policy and its durability. Policies that affect long-lived capital operate in a political environment in which future governments can extend, modify, or repeal them. When investment is irreversible and costly to shift across time, the timing and persistence of policy can therefore matter independently of its expected fiscal size. This issue extends beyond renewable subsidies to other policies intended to redirect long-lived private investment.

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A Additional model details

A.1 Subsidy eligibility

The wind and solar eligibility indicators are

$$\chi_t^W = \mathbf{1}[s_t = 1], \quad \chi_t^S = \mathbf{1}[t < 2028] + \mathbf{1}[t \geq 2028]\mathbf{1}[s_t = 1]. \quad (\text{A.1})$$

Wind invested while the subsidy is active enters the eligible vintage $K_{jt}^{W,PTC}$ according to (4.4). The production tax credit is grandfathered. Solar receives the investment tax credit at construction.

A.2 Electricity prices and profits

Substituting the Pareto survival function (4.9) into the expected-price expression (4.11) gives the wholesale price in closed form. Define

$$x_{jt}^{d2} = Q_{jt}^0, \quad x_{jt}^{d3} = Q_{jt}^0 + K_{jt}^{\text{lo}}, \quad \bar{Q}_{jt} = Q_{jt}^0 + K_{jt}^{\text{lo}} + K_{jt}^{\text{hi}}.$$

Then

$$\begin{aligned} \bar{p}_{jt}^E &= mc_{jt}^{\text{lo}} \left(\frac{x_{jt}^{\text{min}}}{x_{jt}^{d2}} \right)^{\theta_j} \\ &\quad + (mc_{jt}^{\text{hi}} - mc_{jt}^{\text{lo}}) \left(\frac{x_{jt}^{\text{min}}}{x_{jt}^{d3}} \right)^{\theta_j} \\ &\quad + (V_j - mc_{jt}^{\text{hi}}) \left(\frac{x_{jt}^{\text{min}}}{\bar{Q}_{jt}} \right)^{\theta_j}, \end{aligned} \tag{A.2}$$

with each term capped at its coefficient when the corresponding threshold lies below x_{jt}^{min} . Additional zero-marginal-cost capacity raises the dispatch thresholds and lowers the expected wholesale price.

Regional net exports are

$$X_{jt} = G_{jt} - \bar{E}_{jt}.$$

The local price is related to the interconnection hub price by (4.12). The associated transmission cost,

$$c_j(X_{jt}) = \frac{\gamma}{2} \frac{X_{jt}^2}{\bar{T}\bar{L}_j},$$

is a real resource cost. The congestion basis itself is a transfer and is returned to households through $\mathcal{R}_{jt}$.

Electricity-sector profit is

$$\begin{aligned} \Pi_{jt}^E &= p_{jt}^E [(CF_j^W K_{jt}^W + CF_j^S K_{jt}^S + Q_{jt}^{\text{base}})8760 + \text{gen}_{jt}^G + \text{gen}_{jt}^C] \\ &\quad + \phi^{PTC} CF_j^W 8760 K_{jt}^{W,PTC} - mc_t^G \text{gen}_{jt}^G - mc_t^C \text{gen}_{jt}^C. \end{aligned} \tag{A.3}$$

The first line is electricity-market revenue; the second adds the production tax credit earned by eligible wind and subtracts fossil fuel costs.

A.3 Capacity adequacy and the reserve margin

Firm capacity earns revenue when fossil generators are marginal and during periods of scarcity. I also impose a capacity-adequacy requirement based on the system's reliability target to ensure sufficient investment in firm capacity. Accredited capacity is

$$\mathcal{K}_{jt}^{\text{acc}} = K_{jt}^G + K_{jt}^C + Q_{jt}^{\text{base}} + \zeta_j^W K_{jt}^W + \zeta_j^S K_{jt}^S, \quad (\text{A.4})$$

where ζ_j^W and ζ_j^S are the effective load-carrying capability (ELCC) of wind and solar. The system must satisfy

$$\Pr(D_{jt}^{\text{fc}} > \mathcal{K}_{jt}^{\text{acc}}) \leq \varepsilon, \quad (\text{A.5})$$

where D_{jt}^{fc} is exogenous forecast load and ε is the loss-of-load-probability target. The requirement is based on exogenous forecast demand, so it does not respond to the model's endogenous electricity price or demand. Under the Pareto load distribution, it gives a minimum level of accredited capacity. When existing capacity falls below this level, gas fills the shortfall. The multiplier on (A.5) is the capacity price p_{jt}^{cap} .

When the constraint binds,

$$p_{jt}^{\text{cap}} = \max \left\{ p_t^G(r_t^K + \delta^G) - \text{MPK}_{jt}^{G,\text{energy}}, 0 \right\}. \quad (\text{A.6})$$

The capacity price is therefore the “missing money” between the user cost of gas capacity and its return in the energy market. It is a shadow value and does not enter aggregate welfare. I size each region's adequacy requirement to its own forecast demand.

A.4 Formal equilibrium definition

Within-period equilibrium. Given generation-capital stocks $\mathbf{K}_{jt}$, policy state s_t , and the exogenous demand and cost paths, a within-period equilibrium consists of regional electricity prices $\{p_{jt}^E\}$, interconnection hub prices $\{\bar{p}_{mt}^E\}$, generation and dispatch, household and firm electricity demands, and net trade such that:

1. each region dispatches generation in merit order and the wholesale price satisfies (4.11);
2. household and firm electricity demands satisfy (4.10);
3. local and hub prices satisfy (4.12);
4. electricity markets clear within each interconnection; and
5. congestion-basis payments are returned through $\mathcal{R}_{jt}$.

Dynamic equilibrium. A dynamic equilibrium is a within-period equilibrium in every period together with capital stocks, investment policies, financial wealth, consumption, and the national capital rental such that:

1. generation investment solves (4.5) subject to irreversibility, capital accumulation (4.3), and the wind subsidy vintage (4.4);
2. capital evolves according to its laws of motion;
3. the stochastic discount factor is consistent with household consumption;
4. the national production-capital market satisfies (4.8);
5. regional labor markets and the aggregate goods market clear;
6. the federal government budget balances;
7. the policy state follows Π ; and
8. the economy converges to the terminal stochastic steady state.

The federal government finances subsidy outlays through the labor-income tax and national lump-sum transfers,

$$\sum_j \varsigma_t w_{jt} L_{jt} = \Phi_t + \sum_j N_j T_t^{nat}, \quad (\text{A.7})$$

where Φ_t denotes aggregate subsidy expenditure.

A.5 Resource accounting

By aggregate goods-market clearing, the consumption component of welfare in (4.14) can equivalently be written

$$W^C = \sum_t \beta^t \left[\sum_j Y_{jt} - \sum_{j,k} p_t^k I_{jt}^k - \sum_{j,k} \Psi^k(I_{jt}^k, K_{jt}^k) - \sum_j (mc_t^G gen_{jt}^G + mc_t^C gen_{jt}^C) - \sum_j c_j(X_{jt}) - \Delta K_{P,t}^{\text{total}} \right]. \quad (\text{A.8})$$

Generation investment, adjustment costs, fossil fuel use, transmission, and production-capital investment are real resource costs. Federal subsidies, the taxes that finance them, household electricity payments, and congestion-basis rebates are transfers and therefore cancel in aggregate welfare. Generation investment is valued at its full resource price p_t^k , rather than the ITC-reduced private price $\tilde{p}_t^k$, so that the ITC remains a transfer rather than appearing as a reduction in resource cost.

B Computational algorithm

The model is a non-stationary stochastic spatial general equilibrium. I solve it by nesting the regional dynamic investment problems inside the within-period electricity-market and general-equilibrium fixed points.

B.1 Terminal stochastic steady state

Backward induction requires a terminal condition. I solve the stationary value function at terminal fundamentals by value-function iteration on a three-dimensional capital grid (K^S, K^W, K^G) crossed with the two policy states. At each grid point, the three investment choices $(I^S, I^W, I^G) \geq 0$ are optimized by coordinate-ascent golden-section search.

The eligible wind vintage $K^{W,PTC}$ is not an additional grid dimension. Conditional on eligibility, the production credit is separable from the energy value of wind, so its contribution

can be represented by the present value

$$q^{W,PTC} = \frac{\phi^{PTC} C F_j^W \text{HRS}}{1 - \beta(1 - \delta^W)(1 - h^{exit})}. \quad (\text{B.1})$$

This value enters the marginal return to wind investment only when new capacity qualifies for the subsidy.

B.2 Backward-induction transition

Given the terminal value function, I solve the finite-horizon transition from 2010 to 2130 backward with time-varying fundamentals; quantitative results are reported through 2070.

As a numerical check, I independently verify the resulting investment policies against the Euler conditions using a marginal-value (q -method) solver. At the optimum, Euler residuals are approximately 10^{-7} , the interpolation-error floor.

B.3 Within-period and general-equilibrium fixed points

Within each period, a low-dimensional fixed point jointly determines price-elastic electricity demand, merit-order prices, and the scale of the Pareto load distribution. Interconnection hub prices clear net trade among member regions. An outer fixed point jointly determines state-contingent local electricity prices, the national capital rental, and aggregate consumption.

C Illustration of the investment mechanism

I solve the full model for the Eastern Interconnection along a deterministic policy path in which the subsidy is active, lapses for six years, and is then permanently restored. Figure 11 follows the Midwest, where federal support is important for marginal wind investment.

While the subsidy is active, Midwest wind investment runs at approximately 11 GW per year. When the subsidy lapses, newly built wind no longer qualifies for the PTC and wind investment falls to zero.

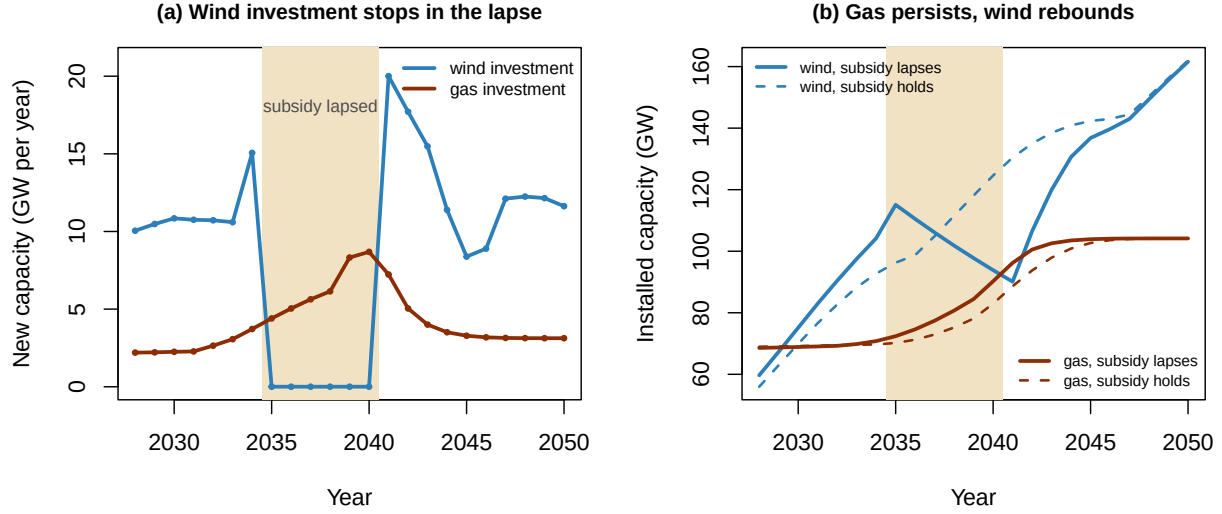


Figure 11: Investment and capacity around a temporary subsidy lapse.
Notes. Full model solved for the Eastern Interconnection along a deterministic policy path in which the subsidy is active, lapses over 2035–2040, and is then restored (the Midwest).

D Policy-process calibration

This appendix reports the pre-2028 policy beliefs and the political transition matrix used to construct the policy process in Section 5.3.

D.1 Pre-2028 policy path

Table 8: Common pre-2028 policy path.

Years	Belief	Pr(lapse active)	Event
2010	active	—	initial state $s_0 = 1$
2010–2021, off-cliff	active, low lapse risk	0.08	
2012, 2014, 2016, 2019, 2021	scheduled expiration	0.60	PTC expiration dates
2022–2027	active	0.00	IRA commitment
2027	lapse in 2028	1.00	OBBBA repeal
2028 onward	regime-specific	—	counterfactual branch

Notes. The pre-2028 path is common to every counterfactual. The counterfactual policy processes begin in 2028.

D.2 Federal political control

I group each Congress from 1993 through 2025 as a Democratic unified control, Republican unified control,, or divided government. Table 9 reports the transition frequencies.

Table 9: Federal political-control transitions, 1993–2025.

Current Congress	Next Congress		
	Democratic trifecta	Republican trifecta	Divided
Democratic trifecta	0.00	0.00	1.00
Republican trifecta	0.00	0.33	0.67
Divided	0.20	0.30	0.50

Notes. Transition frequencies are calculated across two-year Congresses. The ergodic frequencies of Democratic trifecta, Republican trifecta, and divided government are 0.12, 0.27, and 0.61, respectively.

E Calibration and data

Table 10 reports regional summary statistics, and Figure 12 plots the principal exogenous paths.

Table 10: Summary statistics: the U.S. wind, solar, and gas sector by region.

Region	Inter-connect	Wind CF	Solar CF	Wind (GW)	Solar (GW)	Gas (GW)	Demand (TWh)
Southwest	West	0.38	0.32	5.7	7.6	20.7	117
Central	East	0.37	0.26	28.8	0.6	24.8	263
Midwest	East	0.34	0.21	41.3	15.5	98.0	667
Texas	ERCOT	0.34	0.29	42.3	22.5	64.9	403
Northwest	West	0.34	0.29	20.0	12.0	29.9	347
Carolinas	East	0.28	0.25	0.4	8.4	25.6	214
California	West	0.27	0.31	6.5	22.3	34.6	249
Mid-Atlantic	East	0.27	0.19	3.7	11.2	90.3	751
New England	East	0.26	0.19	1.6	3.1	15.8	112
New York	East	0.23	0.19	2.9	2.7	20.7	140
Tennessee	East	0.10	0.25	0.0	0.6	6.9	186
Southeast	East	0.04	0.27	0.0	5.7	35.3	227
Florida	East	0.04	0.27	0.0	10.9	54.6	241
U.S. total				153	123	522	3,919

Notes. Capacity factors are net generation over nameplate capacity, 2019–2024 average (EIA-923/EIA-860); regions with negligible wind show 0. Installed capacity is 2024 (EIA-860). Demand is 2024 electricity sales (AEO 2025). Sources: EIA-860, EIA-923, EIA AEO 2025.

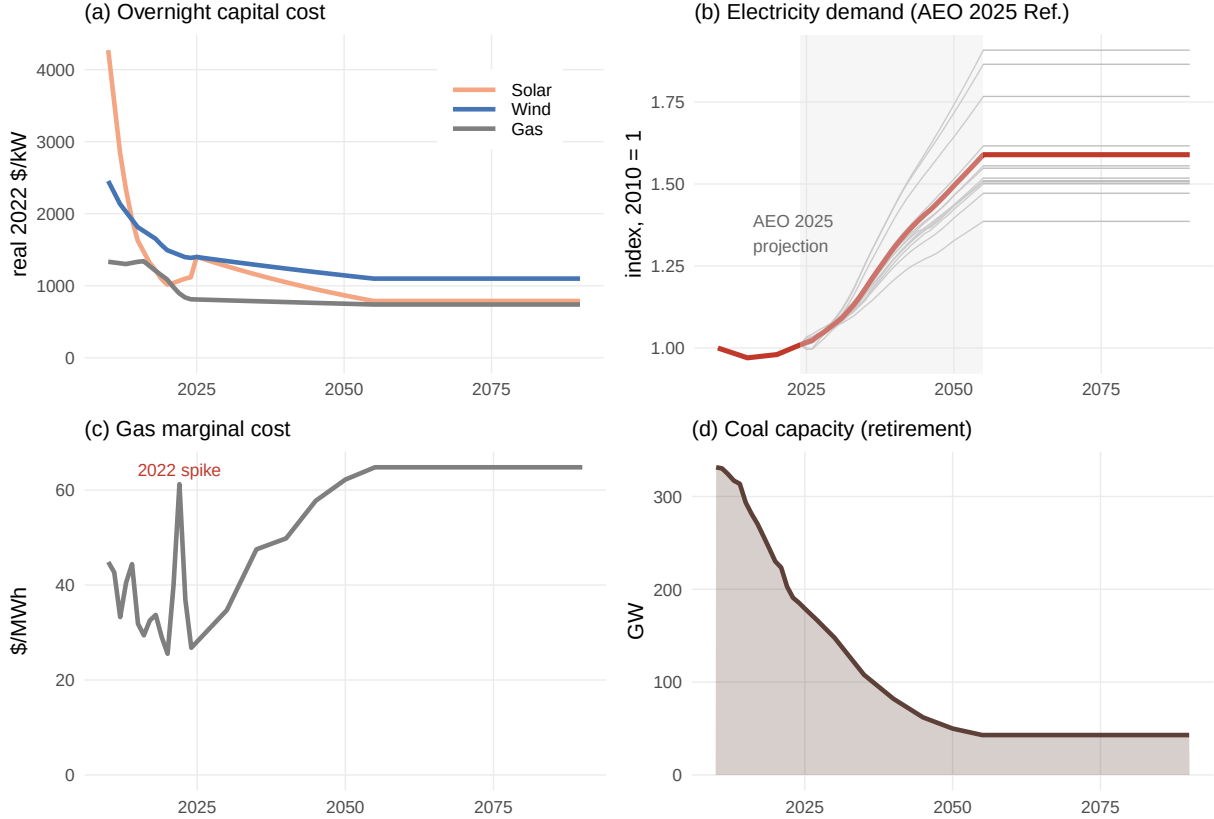


Figure 12: The exogenous processes that drive the transition.

Notes. Panel (a): overnight capital costs (real 2022 \$/kW). Panel (b): electricity-demand index (2010=1). Panel (c): gas marginal cost (\$/MWh). Panel (d): coal capacity (GW). Sources: NREL ATB, LBNL/EIA, and EIA AEO 2025.

E.1 Capacity, generation, and capacity factors

Initial generation stocks and the historical coal-retirement path come from EIA-860 and are aggregated from generators to the 13 model regions by technology. After the observed period, national coal retirements follow the EIA AEO path and are allocated across regions.

Regional capacity factors are calculated as the ratio of EIA-923 net generation by EIA-860 nameplate capacity (averaged over 2019–2024):

$$CF_j^k = \frac{\text{net generation}}{\text{nameplate capacity} \times 8760}.$$

Gas is represented using the combined-cycle capacity factor. Fossil marginal costs combine delivered fuel prices with heat rates and transition toward the AEO 2025 path after the historical period.

E.2 Demand and hourly load

Regional annual electricity demand follows the EIA AEO 2025 Reference case, cross-walked to the 13 model regions. For each region, I estimate the Pareto tail parameter θ_j by maximum likelihood using the upper tail of hourly demand (EIA-930).

E.3 Prices, trade, and emissions

The regional trade slope is

$$s_j = \frac{\gamma}{TL_j}.$$

For regions with observed hub prices, I estimate the slope using the average wholesale-price basis and average net position. For the remaining regions, I impute it from a common congestion parameter γ and observed gross interchange capacity. Net positions and interchange are from EIA-930. Texas is treated as a separate interconnection with no trade with the other model regions.

Emissions factors are $\xi_G = 0.40$ and $\xi_C = 0.95$ tCO₂/MWh from EPA eGRID.

E.4 Federal tax-credit history

Figure 13 summarizes the statutory history used to construct the historical policy path.

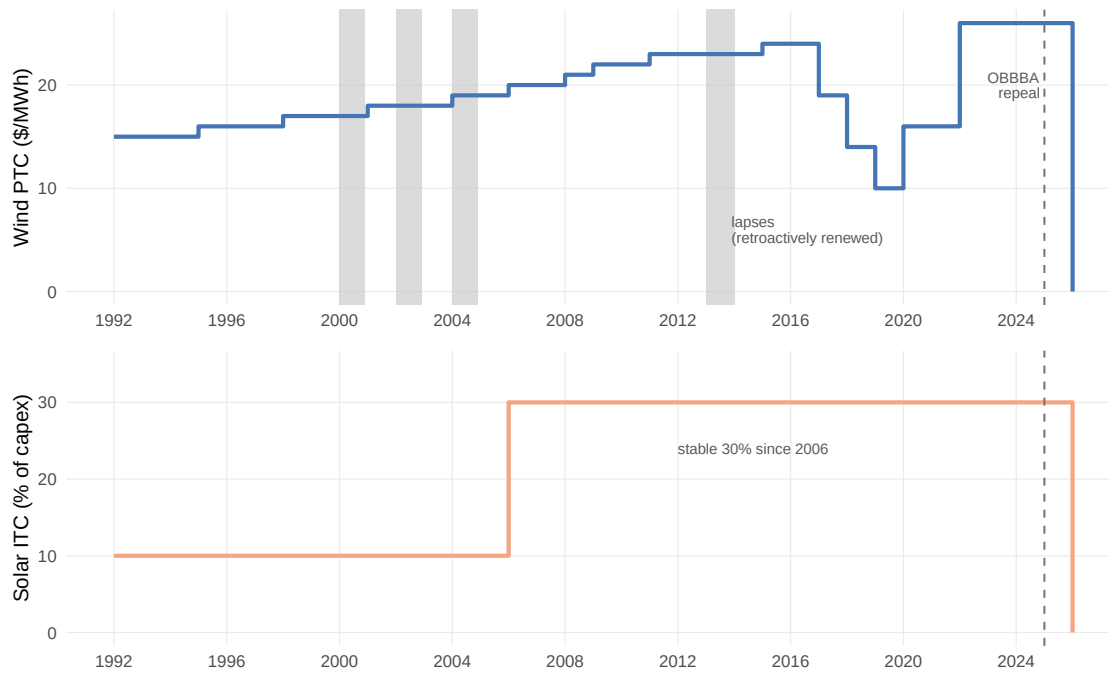


Figure 13: Federal wind and solar tax-credit history, 1990–2025.

Notes. Source: enacting statutes and IRS notices.

Table 11: The production tax credit in the economics of a representative wind project.

	\$/MWh
<i>Revenue per MWh, during the ten-year credit period</i>	
Wholesale energy price	50
Production tax credit (\$45)	26
Total	76
<i>Levelized cost per MWh, unsubsidized</i>	
Capital (\$1,600/kW, 25 yr, 5%)	39
Fixed O&M (\$40/kW-yr)	14
Levelized cost of energy	53
<i>Net profit per MWh</i>	
Without the credit	−3
With the credit	+23
Credit as a share of revenue	34%

Notes. All entries are dollars per megawatt-hour generated.

Table 12: Interconnection trade slope by region.

Region	Interconnection	X_j (GW)	$\overline{TL}_j$ (GW)	s_j	Source
Mid-Atlantic	Eastern	+5.9	7.9	0.24	direct
Central	Eastern	+1.1	1.1	1.79	imputed
Southeast	Eastern	+1.0	2.0	1.00	imputed
New England	Eastern	-0.2	0.2	8.56	imputed
Carolinas	Eastern	-0.3	1.1	1.80	imputed
Tennessee	Eastern	-0.4	3.0	0.68	imputed
Florida	Eastern	-0.4	0.4	4.98	imputed
New York	Eastern	-1.6	2.1	0.99	imputed
Midwest	Eastern	-2.7	6.0	3.58	direct
Northwest	Western	+3.5	3.9	0.52	direct
Southwest	Western	+2.1	4.0	0.52	imputed
California	Western	-6.0	6.0	0.34	imputed
Texas	Texas	-0.2	0.2	12.80	autarky

Notes. The trade slope $s_j = \gamma/\overline{TL}_j$ is the wholesale price basis a region pays per GW of net exports X_j (positive = net exporter). For regions with hub prices (ICE, and NYISO for New York), s_j is estimated *directly* as the mean price basis over the mean net position; for the others it is *imputed* as $\gamma/\overline{TL}_j$ from a common congestion constant $\gamma \approx 2$ (the median of $s_j\overline{TL}_j$ across the priced regions) and the gross-interchange proxy $\overline{TL}_j$. Sources: ICE, NYISO, EIA-930.

F Additional model validation

Section 5.4 reports the principal untargeted moments. This appendix provides the corresponding dynamic and regional comparisons.

F.1 Capacity

Figure 14 compares modeled and observed national wind and solar capacity over 2010–2024 and the 2024 regional cross-section. National wind and solar capacity levels enter the internal calibration; their allocation across regions does not. The model reproduces the regional ordering of wind capacity particularly closely, with a Spearman rank correlation of 0.96, while the corresponding solar correlation is 0.57.

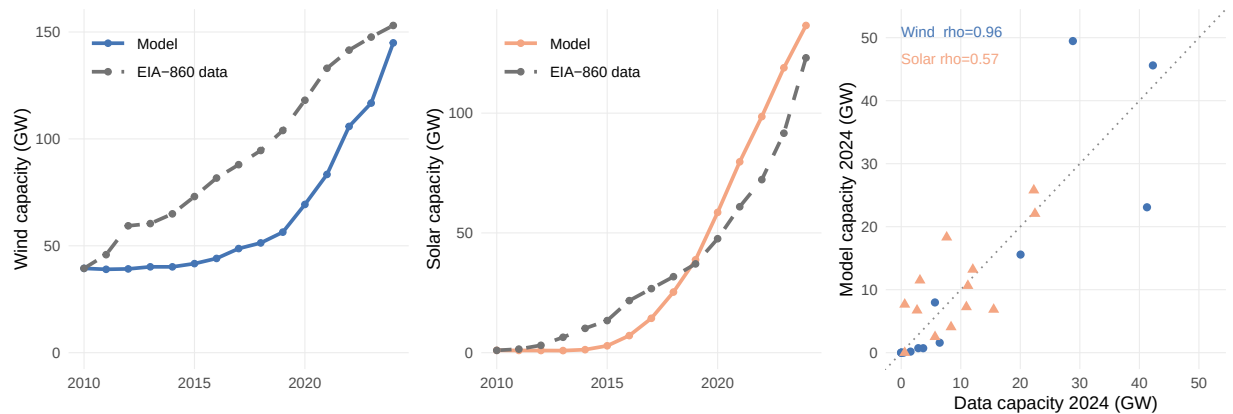


Figure 14: Model versus data: capacity, 2010–2024.

Notes. Panels (a)–(b) compare national wind and solar capacity in the model and EIA-860 data. Panel (c) compares modeled and observed 2024 capacity across the 13 regions. Source: EIA-860.

F.2 Wholesale electricity prices

Figure 15 compares modeled wholesale prices with observed trading-hub prices by interconnection. The model reproduces the broad price level across interconnections and ERCOT as the low-price region.

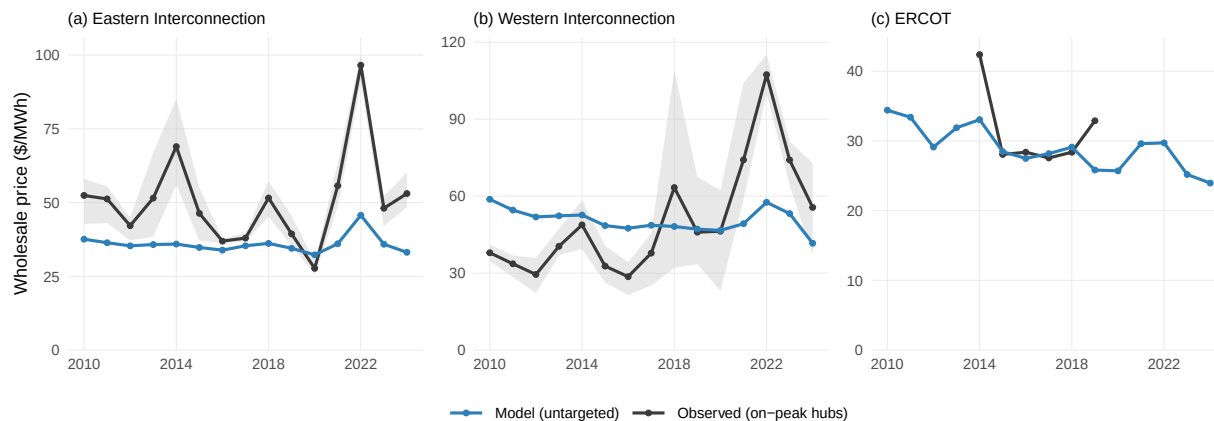


Figure 15: Untargeted wholesale price, by interconnection.

Notes. The black line is the mean across hubs and the grey band is the range. Eastern hubs are PJM West, the Indiana Hub, and the Mass Hub; western hubs are Mid-C, NP15, SP15, and Palo Verde. Source: EIA wholesale market data (ICE hub prices).

G Robustness

G.1 Policy persistence

The expected duration of a lapse is the main uncertainty in the policy calibration. I compare the baseline with a faster renewal regime, in which the subsidy is active 85% of the time in the long run, and with permanent repeal. Table 13 reports the results.

G.2 Climate discount rate

In the baseline, private consumption is discounted at the model's market rate while climate damages are discounted at the social rate used in the EPA SCC calculation. Table 14 varies the rate applied to climate damages. Even when climate damages are discounted at the market rate, the value of permanence remains positive.

G.3 National capital market

The baseline allows the national capital rental r_t^K to clear period by period. I also solve the model holding r_t^K at its terminal modified-golden-rule value. The endogenous rental moves by less than 0.01%, and the counterfactual results are essentially unchanged.

Table 13: Value of a permanent subsidy across all forward regimes (present value, $\text{SCC} = \$190/\text{tCO}_2$).

Permanent subsidy vs.	Climate benefit (\$B)	Consumption cost (\$B, PV)	Net value (\$B)	Breakeven SCC (\$/tCO ₂)
Budget-neutral uncertain subsidy	321	113	+208	67
Realistic uncertain subsidy (post-OBBBA)	340	135	+205	76
No subsidy	502	247	+255	94
Cross-party	48	5	+43	21

Notes. As Table 5, for all four forward regimes, with the breakeven social cost of carbon in the last column. *Realistic uncertain subsidy* is the data-estimated post-OBBBA regime and *No subsidy* the permanent repeal. The value is positive and the breakeven lies far below the EPA’s \$190 in every regime.

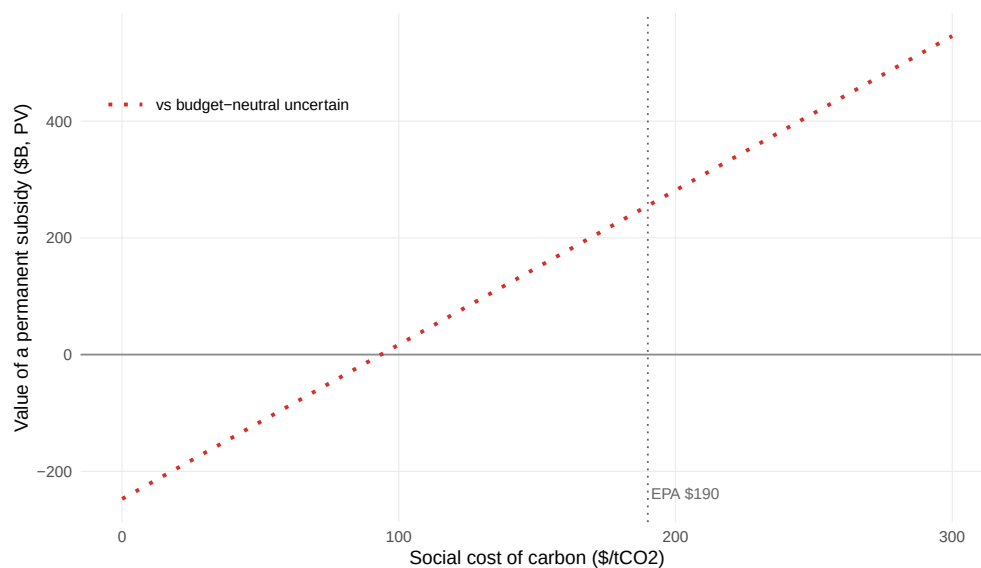


Figure 16: Value of permanence and the social cost of carbon.

Notes. Welfare difference between the permanent and budget-neutral uncertain subsidies as the social cost of carbon varies.

Electricity-generation investment is small relative to the national capital stock, so feedback through the aggregate rental rate is quantitatively minor.

G.4 Perfect-foresight investment by technology

Figure 17 separates the renewable-investment response in Section 7.3 into wind and solar. The sharp anticipatory response is concentrated in wind, for which eligibility determines the stream of production-tax-credit payments associated with newly qualifying capacity.

Table 14: Welfare results across the climate discount rate.

Climate rate	Value of permanence (permanent – uncertain)			Value of the subsidy (uncertain – no subsidy)		
	Climate (\$B)	Value (\$B)	Breakeven SCC	Climate (\$B)	Value (\$B)	Breakeven SCC
1.5%	387	+273	56	221	+87	115
2.0% (baseline)	321	+208	67	181	+47	140
3.0%	224	+110	96	124	–10	205
4.5% (market)	134	+21	161	72	–62	354

Notes. The value of a permanent subsidy over the budget-neutral uncertain subsidy (columns 1–3) and of the uncertain subsidy over no subsidy (columns 4–6), as the rate used to discount the climate benefit varies. Only the climate benefit is re-discounted; the consumption cost is held at its present value at the model rate (\$113 billion for permanence, \$134 billion for the subsidy), since the discount factor β is a structural parameter. The baseline is the EPA’s 2% social rate; the 4.5% row discounts the climate benefit at the model market rate, so the climate externality is treated like private consumption. Value = climate – consumption cost; breakeven SCC is the social cost of carbon at which the value is zero (the reference path is scaled from the EPA 2020 value of \$190/tCO₂). The value of permanence stays positive at every rate; the value of the subsidy turns negative once the climate benefit is discounted at the market rate.

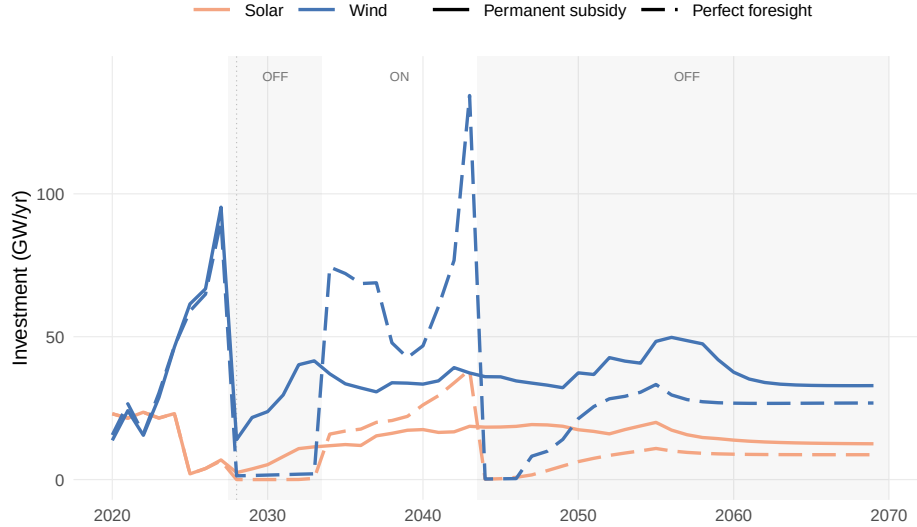


Figure 17: Wind and solar investment under perfect foresight.

Notes. Annual wind and solar investment under permanent support and the deterministic perfect-foresight policy. Shaded areas indicate periods in which the subsidy is unavailable under the foresight path.

H Additional counterfactual results

H.1 Regional results

Figure 18 reports the permanent-minus-uncertain capacity gaps and uncertain-minus-permanent emissions gap for all 13 regions. The largest wind-capacity differences occur in the interior regions where wind resources are strong and federal support affects the marginal investment decision. The gas-capacity and emissions responses need not occur in the same regions as the renewable-investment response because electricity is traded within interconnections.

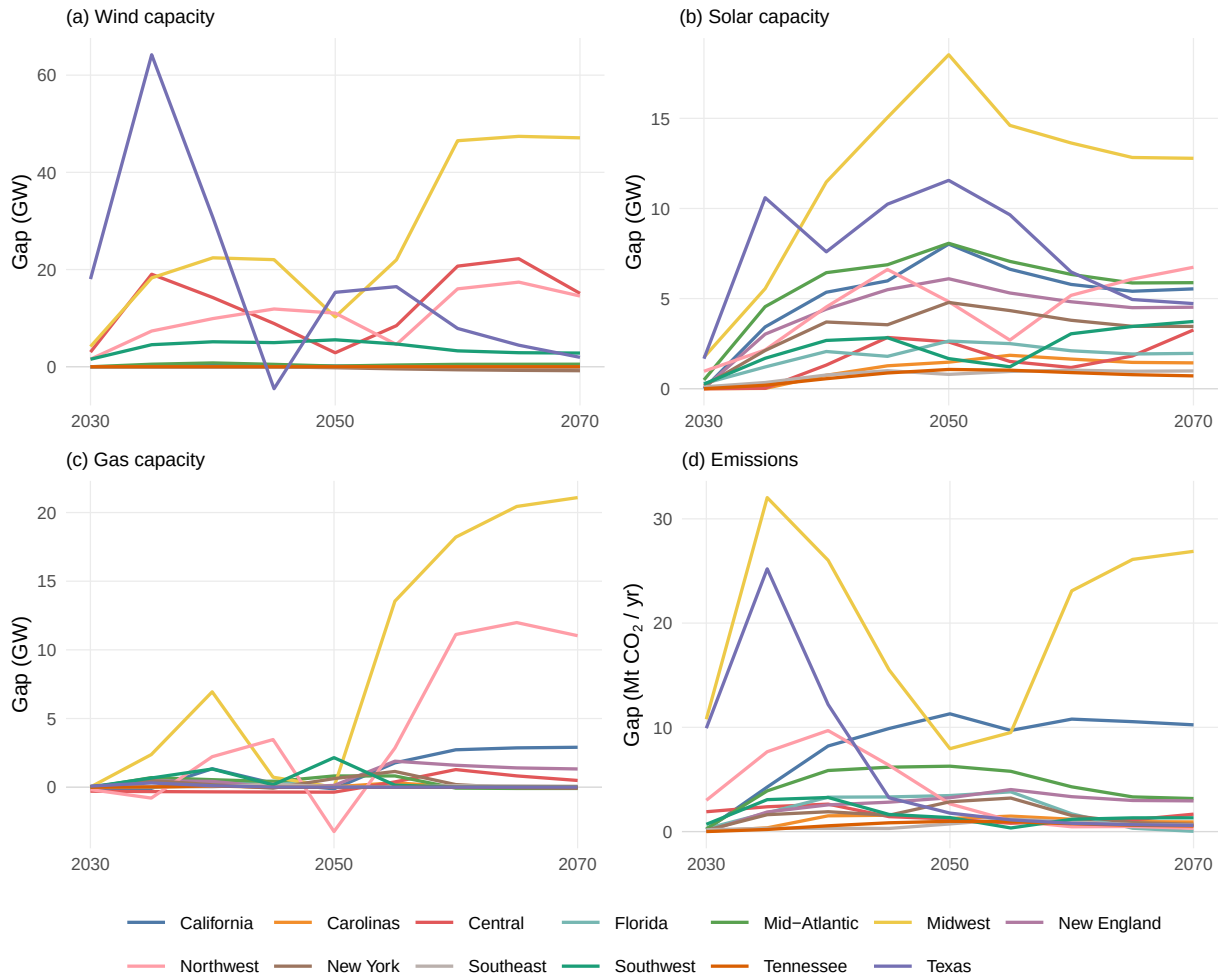


Figure 18: Regional gaps over time, all thirteen regions.

Notes. Gaps between a permanent subsidy and the budget-neutral uncertain subsidy at five-year intervals. Panels (a)–(b) report wind and solar capacity as permanent minus uncertain. Panels (c)–(d) report gas capacity and annual CO₂ emissions as uncertain minus permanent.

Table 15 reports the regional decomposition of the value of the uncertain subsidy relative to no subsidy.

Table 15: The value of the uncertain subsidy by region, decomposed (uncertain versus no subsidy).

Region	Climate gains				Consumption					
	Coal	Ex. gas	New gas	Total	Invest.	Income	Wages	Profit	Total	Value
Midwest	50	16	7	73	-45	29	0	0	-15	57
California	0	20	2	21	-16	6	2	-3	-11	11
Mid-Atlantic	0	14	2	16	-10	5	0	0	-6	11
Florida	1	5	0	7	-5	2	0	0	-3	4
Carolinas	4	0	0	5	-2	0	0	0	-2	3
New England	0	8	1	9	-9	4	1	-2	-6	3
New York	0	6	0	6	-6	3	1	-1	-4	2
Southeast	1	1	0	2	-1	0	0	0	-1	1
Texas	13	4	0	17	-49	34	-4	1	-18	-1
Southwest	3	2	0	5	-6	-2	1	0	-7	-2
Tennessee	0	2	0	3	-11	1	0	0	-10	-7
Northwest	12	-2	1	12	-20	-8	7	-3	-24	-12
Central	0	2	4	6	-23	-7	3	0	-27	-21
National	85	56	40	181	-203	66	12	-9	-134	47

Notes. Present value in billions of dollars, by region, sorted by value. Climate gains value the region's own avoided CO₂ at the EPA year-specific social cost of carbon discounted at 2%, split by source (existing coal, existing gas, new gas). Consumption is the change in the present value of the region's consumption; income, wages, and profit raise it, generation investment enters negatively. Value = climate gains + consumption. Rows sum to the National row.